

LERAY–HOPF WEAK SOLUTIONS OF THE 2D NAVIER–STOKES EQUATIONS ON THE TORUS: AN EXPOSITORY CONSTRUCTION

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Abstract. We give a self-contained expository construction of Leray–Hopf weak solutions to the two-dimensional incompressible Navier–Stokes equations on the flat torus, via Galerkin approximation, energy estimates, and Aubin–Lions compactness. We then prove that such solutions are unique in two dimensions, and that initial data in V give rise to globally defined strong solutions. A final section carries the same estimates to $\mathbb{T}^3$ and exhibits the precise point at which they fail to close. No originality is claimed; the account is assembled from standard sources and written at the level of a first graduate course.

Key words. Navier–Stokes, Leray–Hopf, weak solutions

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1. Introduction. Whether smooth, globally defined solutions to 3-dimensional Navier–Stokes equations exist for all smooth initial data remains open, and is one of the Clay Millennium Prize problems [5]; in two dimensions the corresponding question was settled affirmatively by Ladyzhenskaya [8]. The obstruction is not a failure of existence but of uniqueness and regularity: Leray’s weak solutions [9], extended to bounded domains by Hopf [7], exist globally in three dimensions, and it is their smoothness and uniqueness that remain unknown.

This paper does not aim to provide a constructive contribution to the frontier of science aimed at finding suitable frameworks to solving Navier–Stokes existence and smoothness problem, nor does it aim to faithfully reconstruct every piece of derivation showcased in prior work. The account follows the treatment of Robinson, Rodrigo and Sadowski [11] most closely, with the functional-analytic background drawn from Brezis [2] and Evans [4], and the Navier–Stokes framework from Temam [12] and Constantin and Foias [3]. Instead, this exposition will be scoped to construct Leray–Hopf weak solutions in 2 dimensions and prove their uniqueness, working on $\mathbb{T}^2 = \mathbb{R}^2/2\pi\mathbb{Z}^2$ throughout. This choice removes the boundary, so no trace theory or boundary regularity is required, and it preserves the compact Sobolev embeddings that the Aubin–Lions argument of section 5 relies on — embeddings unavailable on $\mathbb{R}^2$. The Stokes eigenfunctions are then simply the divergence-free Fourier modes, which makes the Galerkin basis of section 4 explicit. None of these simplifications affect the structure of the Leray–Hopf construction.

2. Standing propositions. The following propositions have to be established to lay out the groundwork for the derivation; each is referenced in its respective placement of use.

DEFINITION 2.1. *If X is a Banach space, we say that $u \in L^1(0, T; X)$ has weak time derivative $\partial_t u \in L^1(0, T; X)$ if*

$$\int_0^T u(s) \partial_t \phi(s) ds = - \int_0^T \partial_t u(s) \phi(s) ds$$

for all $\phi \in C_c^\infty(0, T)$.

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LEMMA 2.2. Suppose that $u, g \in L^1(0, T; X)$. Then the following are equivalent:

- (i) $\partial_t u = g$ in the sense of Definition 2.1;
- (ii) there exists $\zeta \in X$ such that

$$u(t) = \zeta + \int_0^t g(s) ds \text{ for almost every } t \in [0, T];$$

- (iii) for every $f \in X^*$

$$\frac{d}{dt} \langle f, u \rangle = \langle f, g \rangle$$

where the time derivative is taken in a weak sense.

In particular, if (i)-(iii) hold, then u is almost everywhere equal to a continuous function from $[0, T]$ into X .

LEMMA 2.3 (Grönwall's inequality, differential form [4]). Let $\phi : [0, T] \rightarrow \mathbb{R}$ be a non-negative, absolutely continuous function, and let $g \in L^1(0, T)$ be a non-negative function. If ϕ satisfies the differential inequality:

$$\phi'(t) \leq g(t)\phi(t) \text{ for a.e. } t \in (0, T)$$

then for all $t \in [0, T]$,

$$\phi(t) \leq \phi(0) \exp \left(\int_0^t g(s) ds \right)$$

DEFINITION 2.4. The space $\dot{L}^2(\mathbb{T}^2)$ consists of all functions $u \in L^2(\mathbb{T}^2)$ that have zero mean,

$$\int_{\mathbb{T}^2} u(x) dx = 0$$

DEFINITION 2.5 (Notation). To simplify the notation for the reader, we write vector-valued Lebesgue spaces as

$$(2.1) \quad \mathbb{L}^p = [L^p]^2 = L^p \times L^p, \quad 1 \leq p \leq \infty,$$

and, similarly, vector-valued Sobolev spaces as

$$(2.2) \quad \mathbb{H}^s = [H^s]^2 = H^s \times H^s, \quad s \geq 0.$$

Throughout, an unadorned L^p or H^s therefore denotes a space of scalar-valued functions, and its blackboard counterpart the corresponding space of vector fields. Norms of gradients are understood componentwise: for a vector field u we write $\|\nabla u\|_{\mathbb{L}^2}^2 = \sum_{i,j} \|\partial_j u_i\|_{L^2}^2$, and likewise $\|\nabla g\|_{L^2}$ for a scalar g .

We write $\dot{\mathbb{Z}}^2 = \mathbb{Z}^2 \setminus \{0\}$ for the punctured frequency lattice, and $C_{\sigma}^{\infty}(\mathbb{T}^2)$ for the smooth, mean-zero, divergence-free vector fields on $\mathbb{T}^2$.

Analogous to Definition 2.4, we define homogeneous spaces $\dot{\mathbb{L}}^p$ and $\dot{\mathbb{H}}^s$ as consisting of vector-valued functions of $\mathbb{L}^p$ and $\mathbb{H}^s$ spaces respectively, satisfying the zero-mean condition as we will add said condition when considering Navier–Stokes equations in this setting.

The mean is conserved by the flow, so one normalizes this away. This enables us to use Poincaré inequality available on $\mathbb{T}^2$.

THEOREM 2.6 (Parseval identity). *Let $u \in L^2(\mathbb{T}^2)$ have Fourier coefficients*

$$\hat{u}(k) = \frac{1}{(2\pi)^2} \int_{\mathbb{T}^2} u(x) e^{-ik \cdot x} dx, \quad u(x) = \sum_{k \in \mathbb{Z}^2} \hat{u}(k) e^{ik \cdot x}.$$

Then

$$\|u\|_{L^2(\mathbb{T}^2)}^2 = (2\pi)^2 \sum_{k \in \mathbb{Z}^2} |\hat{u}(k)|^2.$$

This normalisation of $\hat{u}$ is used throughout; we write $\hat{u}_k$ for $\hat{u}(k)$ interchangeably.

DEFINITION 2.7. *Let $s \geq 0$. The Sobolev space $H^s(\mathbb{T}^2)$ is defined by*

$$H^s(\mathbb{T}^2) := \{u \in L^2(\mathbb{T}^2) : \|u\|_{H^s}^2 < \infty\}$$

where

$$\|u\|_{H^s}^2 := (2\pi)^2 \sum_{k \in \mathbb{Z}^2} (1 + |k|^2)^s |\hat{u}(k)|^2,$$

normalised so that $\|u\|_{H^0} = \|u\|_{L^2}$, in agreement with Theorem 2.6. We write $\dot{H}^s$ for the corresponding homogeneous norm, in which the weight $(1 + |k|^2)^s$ is replaced by $|k|^{2s}$ and the sum omits the zero mode:

$$\|u\|_{\dot{H}^s}^2 := (2\pi)^2 \sum_{k \in \dot{\mathbb{Z}}^2} |k|^{2s} |\hat{u}(k)|^2 = \|(-\Delta)^{s/2} u\|_{L^2}^2.$$

On $\dot{H}^s(\mathbb{T}^2)$ the two norms are equivalent, since $|k| \geq 1$ for every $k \in \dot{\mathbb{Z}}^2$. Where an identity requires the homogeneous norm exactly, we write $\dot{H}^s$ (resp. $\dot{\mathbb{H}}^s$) rather than H^s .

THEOREM 2.8 (Helmholtz–Leray decomposition on $\mathbb{T}^2$). *The space $\dot{\mathbb{L}}^2$ can be written as*

$$\dot{\mathbb{L}}^2 = H \oplus G$$

i.e. every function $u \in \dot{\mathbb{L}}^2$ can be written in a unique way as

$$u = h + \nabla g,$$

where the vector-valued function h belongs to H and the scalar function g belongs to $\dot{H}^1$; the spaces H of divergence-free fields and G of gradients are those of Definition 2.10 and Definition 2.13 below. Moreover,

(i) the functions h and ∇g are orthogonal in $\dot{\mathbb{L}}^2$,

$$\langle h, \nabla g \rangle = 0,$$

and,

(ii) if in addition u belongs to $\dot{\mathbb{H}}^s$, $s > 0$, then $h \in \dot{\mathbb{H}}^s$ and $g \in \dot{H}^{s+1}$.

Proof. For $k \neq 0$, we define $\mathbb{C}^2 = \text{span}(k) \oplus k^\perp$. Then $\widehat{\mathbb{P}}(k) = I - k \otimes k / |k|^2$ is the orthogonal projection onto $k^\perp$, while $I - \widehat{\mathbb{P}}(k)$ projects onto $\text{span}(k)$. $k \cdot \hat{u}(k) = 0$ (incompressibility) characterizes divergence-free fields, while $\hat{u}(k) \parallel k$ characterizes gradients — applying this at every frequency and invoking Theorem 2.6, we get the orthogonal splitting.

The projection is a Fourier multiplier with $|\widehat{\mathbb{P}}(k)| \leq 1$:

$$|\hat{h}_k| = |\widehat{\mathbb{P}}(k) \hat{u}_k| \leq |\hat{u}_k| \implies \sum_k |k|^{2s} |\hat{h}_k|^2 \leq \sum_k |k|^{2s} |\hat{u}_k|^2 < \infty$$

An implication is that $h \in \dot{\mathbb{H}}^s$ and $\nabla g \in \dot{\mathbb{H}}^s$. It follows therefore that $g \in \dot{H}^{s+1}$. $\square$

DEFINITION 2.9 (Leray projector). *Operator $\mathbb{P}$, commonly referred to as the Leray projector, maps the given vector onto its divergence-free component. We define this transformation as follows. For:*

$$u = \nabla q + v, \text{ with } \nabla \cdot v = 0$$

By Theorem 2.8, such decomposition exists and is unique. We define the unique projection of u onto its divergence-free component (that is, H) as:

$$\mathbb{P}u = v.$$

For the zero mode $\hat{u}(0)$, we mention that on $\dot{\mathbb{L}}^2(\mathbb{T}^2)$ the two standard conventions for $\widehat{\mathbb{P}u}(0)$ coincide, thus we take $\widehat{\mathbb{P}u}(0) = 0$.

DEFINITION 2.10. *We define the space $H = H(\mathbb{T}^2)$ as*

$$H(\mathbb{T}^2) = \{u \in \dot{\mathbb{L}}^2 : u = \sum_{k \in \dot{\mathbb{Z}}^2} \hat{u}_k e^{ik \cdot x}, \hat{u}_{-k} = \bar{\hat{u}}_k \text{ and } k \cdot \hat{u}_k = 0 \ \forall k \in \dot{\mathbb{Z}}^2\}$$

DEFINITION 2.11. *A function $u \in \mathbb{L}^1(\mathbb{T}^2)$ is weakly divergence-free if*

$$\langle u, \nabla \phi \rangle = 0 \text{ for every } \phi \in C^\infty(\mathbb{T}^2)$$

LEMMA 2.12. *Each $u \in H(\mathbb{T}^2)$ is weakly divergence-free and moreover*

$$\langle u, \nabla \phi \rangle = 0 \ \forall \phi \in H^1(\mathbb{T}^2)$$

Proof. If $u \in H(\mathbb{T}^2)$ admits a form:

$$u(x) = \sum_{m \in \dot{\mathbb{Z}}^2} \hat{u}_m e^{im \cdot x}$$

for any $\phi \in C^\infty(\mathbb{T}^2)$, we have the following:

$$\langle u, \nabla \phi \rangle = (2\pi)^2 \sum_{k \in \dot{\mathbb{Z}}^2} \hat{u}_k \cdot \overline{ik \hat{\phi}_k} = 0$$

since $\hat{u}_k \cdot k = 0$ is characterized by the incompressibility constraint. □

DEFINITION 2.13. *The space G on the torus $\mathbb{T}^2$ is defined as:*

$$G(\mathbb{T}^2) = \{u \in \dot{\mathbb{L}}^2 : u = \nabla g, \text{ where } g \in \dot{H}^1(\mathbb{T}^2)\}$$

We assume here that g belongs to $\dot{H}^1$ (rather than to H^1) in order to obtain the uniqueness of g .

COROLLARY 2.14. *The spaces H and G are orthogonal, that is*

$$\langle h, \nabla g \rangle = 0$$

for every $h \in H$ and $\nabla g \in G$.

Proof. Apply Lemma 2.12 with $\phi = g$. □

DEFINITION 2.15. *We define the space of time-independent test functions*

$$V = \{u \in \dot{H}^1(\mathbb{T}^2) : k \cdot \hat{u}_k = 0 \ \forall k\}$$

with norm given as $\|\nabla u\|_{\mathbb{L}^2}$. We additionally define V' as:

$$V' = \{f : k \cdot \hat{f}_k = 0, \sum_{k \neq 0} |k|^{-2} |\hat{f}_k|^2 < \infty\}.$$

with a norm given as:

$$\|f\|_{V'} = \sup_{\|x\|_V \leq 1} |\langle f, x \rangle|$$

By the Riesz representation theorem we identify H with its dual H' , so that

$$\langle f, v \rangle_{H', H} = \langle f, v \rangle_{\mathbb{L}^2}.$$

$V \subset H$ densely and continuously, hence $H \cong H' \subset V'$, and the $V' - V$ pairing extends the $\mathbb{L}^2$ inner product. We do not identify V with V' .

DEFINITION 2.16. *We define the trilinear form as:*

$$b(u, v, w) = \langle (u \cdot \nabla) v, w \rangle$$

where u, v, w are vectors.

THEOREM 2.17 (Lions–Magenes [10, 12]). *Let $V \subset H \cong H' \subset V'$ be a Gelfand triple with $V \hookrightarrow H$ dense and continuous. Suppose $u \in L^2(0, T; V)$ and $u' \in L^2(0, T; V')$, where u' is the distributional time derivative. Then, after modification on a null set, $u \in C([0, T]; H)$, the map $t \mapsto \|u(t)\|_H^2$ is absolutely continuous, and*

$$\frac{d}{dt} \|u(t)\|_H^2 = 2 \langle u'(t), u(t) \rangle_{V', V} \quad \text{for a.e. } t \in (0, T).$$

THEOREM 2.18 (Ladyzhenskaya). *On the periodic torus $\mathbb{T}^2$, for $u \in \dot{H}^1(\mathbb{T}^2)$, which provides the zero-mean condition as given per Definition 2.4, we have the following:*

$$\|u\|_{\mathbb{L}^4} \leq C \|u\|_{\mathbb{L}^2}^{1/2} \|\nabla u\|_{\mathbb{L}^2}^{1/2}$$

Proof. We first record a one-dimensional bound. Let f be smooth and 2π -periodic in a single variable. For any x and any y ,

$$f(x)^2 = f(y)^2 + \int_y^x \frac{d}{ds} [f(s)]^2 ds = f(y)^2 + \int_y^x 2f(s)f'(s) ds.$$

Averaging over $y \in [0, 2\pi]$ and bounding both terms by their absolute values — the second after enlarging the domain of integration from $[y, x]$ to $[0, 2\pi]$, which can only increase it — gives

$$f(x)^2 \leq \frac{1}{2\pi} \int_0^{2\pi} f(y)^2 dy + 2 \int_0^{2\pi} |f(s)| |f'(s)| ds.$$

Since the right-hand side no longer depends on x , it bounds the maximum:

$$(2.3) \quad \max_x f(x)^2 \leq \frac{1}{2\pi} \|f\|_{L^2}^2 + 2 \int_0^{2\pi} |f(x)| |f'(x)| dx.$$

We emphasise that no zero-mean hypothesis on the individual line is needed here; the mean-zero condition on $\mathbb{T}^2$ does not make each one-dimensional slice mean-zero, so the averaging step above replaces the more familiar argument through the intermediate value theorem. The standard definition of the L^2 norm for a function g over $[0, 2\pi]$:

$$(2.4) \quad \|g\|_{L^2} = \left(\int_0^{2\pi} |g(x)|^2 dx \right)^{1/2}$$

Applying Cauchy–Schwarz inequality to (2.3):

$$\int_0^{2\pi} |f(x)| |f'(x)| dx \leq \left(\int_0^{2\pi} |f(x)|^2 dx \right)^{1/2} \left(\int_0^{2\pi} |f'(x)|^2 dx \right)^{1/2}$$

Which lets us express the final bound as the L^2 norm:

$$\int_0^{2\pi} |f(x)| |f'(x)| dx \leq \|f\|_{L^2} \|\nabla f\|_{L^2}$$

We are ready to estimate $\|f\|_{L^4}$. Write $x = (x_1, x_2)$ and abbreviate $M_1(x_2) = \sup_{x_1} f(x_1, x_2)^2$ and $M_2(x_1) = \sup_{x_2} f(x_1, x_2)^2$. Bounding $f^4 = f^2 \cdot f^2$ by $M_1(x_2)M_2(x_1)$ pointwise and using that the bound is a product of a function of x_2 with a function of x_1 :

$$\|f\|_{L^4}^4 = \int_{\mathbb{T}^2} f^4 \leq \iint M_1(x_2)M_2(x_1) dx_1 dx_2 = \left(\int M_1(x_2) dx_2 \right) \left(\int M_2(x_1) dx_1 \right).$$

Apply (2.3) to each slice and then Cauchy–Schwarz in the remaining variable. For the first factor, the slice is $f(\cdot, x_2)$ and the derivative that appears is $\partial_1 f$, so

$$\int M_1(x_2) dx_2 \leq \frac{1}{2\pi} \|f\|_{L^2}^2 + 2\|f\|_{L^2} \|\partial_1 f\|_{L^2},$$

and symmetrically for the second factor with $\partial_2 f$. Note that each factor is of degree two in f , not four. On $\dot{\mathbb{H}}^1$ the Poincaré inequality bounds $\|f\|_{L^2}$ by $\|\nabla f\|_{L^2}$, so both factors are bounded by $C\|f\|_{L^2} \|\nabla f\|_{L^2}$ and

$$\|f\|_{L^4}^4 \leq C\|f\|_{L^2}^2 \|\nabla f\|_{L^2}^2$$

For $u = (u_1, u_2)$ with $|u|^2 = u_1^2 + u_2^2$, we have (using the fact that $(a+b)^2 \leq 2(a^2 + b^2)$):

$$(2.5) \quad \begin{aligned} \|u\|_{\mathbb{L}^4}^4 &= \int (u_1^2 + u_2^2)^2 \leq 2 \sum_i \int u_i^4 \leq 2C \sum_i \|u_i\|_{L^2}^2 \|\nabla u_i\|_{L^2}^2 \\ &\leq 2C \left(\sum_i \|u_i\|_{L^2}^2 \right) \left(\sum_i \|\nabla u_i\|_{L^2}^2 \right) \end{aligned}$$

All terms are non-negative, adding up to $\|u\|_{\mathbb{L}^2}^2$ and $\|\nabla u\|_{\mathbb{L}^2}^2$. Taking the fourth root of the expression:

$$(2.6) \quad \|u\|_{\mathbb{L}^4} \leq C\|u\|_{\mathbb{L}^2}^{1/2} \|\nabla u\|_{\mathbb{L}^2}^{1/2}$$

The above assumed $u \in C^\infty(\mathbb{T}^2; \mathbb{R}^2)$. Both sides of (2.6) are continuous with respect to the $\dot{\mathbb{H}}^1$ norm, and trigonometric polynomials are dense in $\dot{\mathbb{H}}^1(\mathbb{T}^2)$ by Fourier truncation, so the inequality extends to all $u \in \dot{\mathbb{H}}^1(\mathbb{T}^2)$. $\square$

THEOREM 2.19 (Bolzano–Weierstrass). *Every bounded sequence of real numbers has a convergent subsequence.*

THEOREM 2.20 (Banach–Alaoglu [2]). *If X is a separable Banach space, then any bounded sequence in X^* has a weakly-* convergent subsequence.*

COROLLARY 2.21 (Weak compactness). *If X is a reflexive Banach space, then any bounded sequence in X has a weakly convergent subsequence.*

LEMMA 2.22. *If $\mathcal{H}$ is a Hilbert space and $(x_n) \subset \mathcal{H}$, then $x_n \rightarrow x$ if and only if $x_n \rightarrow x$ and $\|x_n\| \rightarrow \|x\|$.*

COROLLARY 2.23 (Sobolev embeddings [2]). *Let $m \geq 1$ be an integer and let $p \in [1, +\infty)$. We have*

$$W^{m,p}(\mathbb{R}^N) \subset L^q(\mathbb{R}^N), \text{ where } \frac{1}{q} = \frac{1}{p} - \frac{m}{N}, \text{ if } \frac{1}{p} - \frac{m}{N} > 0,$$

$$W^{m,p}(\mathbb{R}^N) \subset L^q(\mathbb{R}^N), \forall q \in [p, +\infty), \text{ if } \frac{1}{p} - \frac{m}{N} = 0,$$

$$W^{m,p}(\mathbb{R}^N) \subset L^\infty(\mathbb{R}^N), \text{ if } \frac{1}{p} - \frac{m}{N} < 0,$$

and all those injections are dense and continuous. Moreover, if $m - \frac{N}{p} > 0$ is not an integer, set

$$k = m - \frac{N}{p} \text{ and } \theta = m - \frac{N}{p} - k \quad (0 < \theta < 1).$$

We have, for all $u \in W^{m,p}(\mathbb{R}^N)$,

$$\|D^\alpha u\|_{L^\infty(\mathbb{R}^N)} \leq C \|u\|_{W^{m,p}(\mathbb{R}^N)} \quad \forall \alpha \text{ with } |\alpha| \leq k$$

and

$$|D^\alpha u(x) - D^\alpha u(y)| \leq C \|u\|_{W^{m,p}(\mathbb{R}^N)} |x - y|^\theta \text{ for a.e. } x, y \in \mathbb{R}^N, \forall \alpha \text{ with } |\alpha| = k.$$

In particular, $W^{m,p}(\mathbb{R}^N) \subset C^k(\mathbb{R}^N)$.

3. Structure of weak solutions. We first define the Stokes operator:

DEFINITION 3.1 (Stokes operator). *The Stokes operator A is defined by*

$$Au = -\mathbb{P}\Delta u$$

with its domain given by

$$D(A) = V \cap \mathbb{H}^2.$$

Now, we establish the correspondence between the Laplacian and the Stokes operator in the periodic torus $\mathbb{T}^2$.

THEOREM 3.2 (Laplacian and Stokes operator). *On the torus $\mathbb{T}^2$ we have*

$$-\mathbb{P}\Delta u = -\Delta \mathbb{P}u$$

$\forall u \in \mathbb{H}^2$. Hence,

$$Au = -\Delta u$$

$\forall u \in D(A)$ on $\mathbb{T}^2$.

Proof. Leray projection is a matrix operation acting on $\hat{u}(k)$:

$$\widehat{\mathbb{P}u}(k) = \hat{u}(k) - \frac{k(k \cdot \hat{u}(k))}{|k|^2}, \quad k \neq 0$$

with $\widehat{\mathbb{P}u}(0) = 0$. Whereas the Laplacian acting on u is a scalar operation:

$$-\widehat{\Delta u}(k) = |k|^2 \hat{u}(k).$$

Both operations hence commute due to their respective symbols commuting as a result of scalar multiplication (which is commutative). As Leray projection is applied on a function that was defined on a divergence-free space, it acts as the identity:

$$-\Delta \mathbb{P}u = -\Delta u \quad \square$$

From Theorem 3.2, it follows that the Stokes operator on the torus $\mathbb{T}^2$ is given by the explicit formula

$$(3.1) \quad Au(x) = \sum_{k \in \dot{\mathbb{Z}}^2} |k|^2 \hat{u}_k e^{ik \cdot x}, \quad u \in D(A).$$

Additionally, we provide a regularity estimate:

THEOREM 3.3. *On the torus $\mathbb{T}^2$ we have*

$$\|u\|_{\dot{\mathbb{H}}^{m+2}} = \|Au\|_{\dot{\mathbb{H}}^m}$$

$\forall u \in V$ such that $Au \in \dot{\mathbb{H}}^m$ and all $m \in \mathbb{N}$. In particular, every second derivative is controlled by the Stokes operator:

$$\|\partial_i \partial_j u\|_{\mathbb{L}^2} \leq \|Au\|_{\mathbb{L}^2} \quad \forall u \in D(A), \quad i, j = 1, 2.$$

Proof. On the periodic torus, we use the explicit Stokes formula:

$$\|Au\|_{\dot{\mathbb{H}}^m}^2 = \left\| \sum_{k \in \dot{\mathbb{Z}}^2} |k|^2 \hat{u}_k e^{ik \cdot x} \right\|_{\dot{\mathbb{H}}^m}^2$$

Open the $\mathbb{L}^2$ product with $v = \sum_{k \in \dot{\mathbb{Z}}^2} \hat{v}_k e^{ik \cdot x}$ and $\hat{v}_k \in \mathbb{C}^2$ using the Parseval identity (Theorem 2.6):

$$\|v\|_{\mathbb{L}^2}^2 = \int_{\mathbb{T}^2} v \cdot \bar{v} \, dx = \sum_k \sum_l \hat{v}_k \cdot \bar{\hat{v}}_l \int_{\mathbb{T}^2} e^{i(k-l) \cdot x} \, dx = (2\pi)^2 \sum_k |\hat{v}_k|^2$$

To combine this with derivatives, we invoke the definition of Sobolev spaces:

$$\begin{aligned} \|v\|_{\dot{\mathbb{H}}^m}^2 &:= \|(-\Delta)^{m/2} v\|_{\mathbb{L}^2}^2 \\ (-\widehat{\Delta})^{m/2} v(k) &= |k|^m \hat{v}_k \\ \|v\|_{\dot{\mathbb{H}}^m}^2 &= (2\pi)^2 \sum_k |k|^{2m} |\hat{v}_k|^2 \end{aligned}$$

Starting from $u(x) = \sum_k \hat{u}_k e^{ik \cdot x}$, we find the derivative twice:

$$\Delta u = \sum_{j=1}^2 \partial_j^2 u = \sum_k \left(\sum_{j=1}^2 (ik_j)^2 \right) \hat{u}_k e^{ik \cdot x} = \sum_k (-k_1^2 - k_2^2) \hat{u}_k e^{ik \cdot x} = - \sum_k |k|^2 \hat{u}_k e^{ik \cdot x}.$$

We get:

$$-\Delta u = \sum_k |k|^2 \hat{u}_k e^{ik \cdot x}$$

By definition of the Stokes operator, $A = -\mathbb{P}\Delta$, the symbol $\widehat{\mathbb{P}}(k)$ is the orthogonal projection of $\mathbb{C}^2$ onto $k^\perp$. Since $\hat{u}_k$ lies in $k^\perp$, $\widehat{\mathbb{P}}(k)$ performs an identity operation. Hence, only Laplacian Δ is the performed modification. Hence:

$$Au = \sum_k |k|^2 \hat{u}_k e^{ik \cdot x}$$

Finally, by setting $v := Au$, we match the coefficients exactly and get:

$$v = \sum_k |k|^2 \hat{u}_k e^{ik \cdot x}$$

Matching this against knowing that $v = \sum_{k \in \mathbb{Z}^2} \hat{v}_k e^{ik \cdot x}$ and combining the terms:

$$\begin{aligned} \|Au\|_{\mathbb{H}^m}^2 &= (2\pi)^2 \sum_k |k|^{2m} |\hat{v}_k|^2 = (2\pi)^2 \sum_k |k|^{2m} |k|^4 |\hat{u}_k|^2 \\ &= (2\pi)^2 \sum_k |k|^{2(m+2)} |\hat{u}_k|^2 = \|u\|_{\mathbb{H}^{m+2}}^2. \end{aligned}$$

For the second claim, $\widehat{\partial_i \partial_j u}(k) = -k_i k_j \hat{u}_k$, and $|k_i k_j| \leq |k|^2$ for every k . Parseval therefore gives

$$\|\partial_i \partial_j u\|_{\mathbb{L}^2}^2 = (2\pi)^2 \sum_k |k_i k_j|^2 |\hat{u}_k|^2 \leq (2\pi)^2 \sum_k |k|^4 |\hat{u}_k|^2 = \|Au\|_{\mathbb{L}^2}^2$$

for $u \in D(A)$. □

THEOREM 3.4. *Let Ω be the torus $\mathbb{T}^2$. Then, there exists a family of functions $\mathcal{N} = \{a_1, a_2, a_3, \dots\}$ such that*

(i) $\mathcal{N}$ is an orthonormal basis in $H(\Omega)$;

(ii) $a_j \in D(A) \cap C^\infty(\Omega)$ are eigenfunctions of the Stokes operator, that is $Aa_j = \lambda_j a_j$ for all $j \in \mathbb{N}$ with

$$0 < \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \dots \leq \lambda_j \leq \dots \quad \text{and } \lambda_j \rightarrow \infty; \quad \text{and}$$

(iii) $\mathcal{N}$ is an orthogonal basis in $V(\Omega)$, with $\|a_j\|_V^2 = \lambda_j$.

Proof. Take any $u \in H$ and open it as a sum of its Fourier coefficients $\hat{u}_k$, $k \in \dot{\mathbb{Z}}^2$ with $\hat{u}_k \cdot k = 0$.

$$\begin{aligned} u(x) &= \sum_{k \in \dot{\mathbb{Z}}^2} \hat{u}_k e^{ik \cdot x} \\ &= \frac{1}{2} \sum_{k \in \dot{\mathbb{Z}}^2} \hat{u}_k e^{ik \cdot x} + \frac{1}{2} \sum_{k \in \dot{\mathbb{Z}}^2} \hat{u}_{-k} e^{-ik \cdot x} \\ &= \frac{1}{2} \sum_{k \in \dot{\mathbb{Z}}^2} \hat{u}_k (\cos(k \cdot x) + i \cdot \sin(k \cdot x)) + \frac{1}{2} \sum_{k \in \dot{\mathbb{Z}}^2} \hat{u}_{-k} (\cos(k \cdot x) - i \cdot \sin(k \cdot x)) \end{aligned}$$

Arrange the terms:

$$\begin{aligned} & \frac{1}{2} \sum_{k \in \dot{\mathbb{Z}}^2} \hat{u}_k (\cos(k \cdot x) + i \cdot \sin(k \cdot x)) + \frac{1}{2} \sum_{k \in \dot{\mathbb{Z}}^2} \hat{u}_{-k} (\cos(k \cdot x) - i \cdot \sin(k \cdot x)) \\ &= \sum_{k \in \dot{\mathbb{Z}}^2} \frac{1}{2} \underbrace{(\hat{u}_k + \hat{u}_{-k})}_{\in \mathbb{R}^2} \cos(k \cdot x) + \sum_{k \in \dot{\mathbb{Z}}^2} \frac{1}{2} \underbrace{i(\hat{u}_k - \hat{u}_{-k})}_{\in \mathbb{R}^2} \sin(k \cdot x) \end{aligned}$$

Considering that the space $k^\perp$ is one-dimensional in two dimensions, the unit vector $m_k = k^\perp/|k|$ is alone and there are no pairs to choose. Considering $\hat{u}_k \in \mathbb{C}^2$, the condition $k \cdot \hat{u}_k = 0$ leaves a one-dimensional complex subspace, which can be expressed by two real terms α_k and β_k .

Because $\cos(k \cdot x) = \cos(-k \cdot x)$ and $\sin(k \cdot x) = -\sin(-k \cdot x)$, the modes attached to k and to $-k$ span the same real two-dimensional space; summing over all of $\dot{\mathbb{Z}}^2$ would therefore count each mode twice. Fix once and for all a half-lattice $\Lambda \subset \dot{\mathbb{Z}}^2$ containing exactly one of $\{k, -k\}$ for each pair, for instance

$$\Lambda = \{k \in \dot{\mathbb{Z}}^2 : k_2 > 0\} \cup \{k \in \dot{\mathbb{Z}}^2 : k_2 = 0, k_1 > 0\}.$$

Then

$$u(x) = \sum_{k \in \Lambda} \alpha_k m_k \cos(k \cdot x) + \sum_{k \in \Lambda} \beta_k m_k \sin(k \cdot x)$$

and the family $\{m_k \cos(k \cdot x), m_k \sin(k \cdot x)\}_{k \in \Lambda}$ is linearly independent. Upon normalization in $\mathbb{L}^2$ it forms an orthonormal basis of H consisting of smooth eigenfunctions of the Stokes operator, each $m_k \cos(k \cdot x)$ and $m_k \sin(k \cdot x)$ having eigenvalue $|k|^2$ by (3.1). Enumerating Λ so that $|k|$ is non-decreasing gives the ordering of the eigenvalues in (ii). Now, let $u, v \in \mathcal{N}$, then we have:

$$\langle \nabla u, \nabla v \rangle = \langle u, -\Delta v \rangle = \langle \mathbb{P}u, -\Delta v \rangle = \langle u, -\mathbb{P}\Delta v \rangle = \langle u, Av \rangle$$

Writing $Av = \lambda_v v$ for the eigenvalue λ_v attached to v , hence:

$$\langle \nabla u, \nabla v \rangle = \langle u, Av \rangle = \lambda_v \langle u, v \rangle$$

So $\mathcal{N}$, orthonormal in H , is orthogonal in V , and $\|a_j\|_V^2 = \langle \nabla a_j, \nabla a_j \rangle = \lambda_j$. It is not orthonormal in V unless every $\lambda_j = 1$, which is why (iii) claims orthogonality only. $\square$

Suppose now that u is a smooth, divergence-free solution of the Navier–Stokes equations:

$$\partial_t u - \nu \Delta u + (u \cdot \nabla)u + \nabla p = 0.$$

We take the $\mathbb{L}^2$ inner product with $\phi \in V$:

$$\langle \partial_t u, \phi \rangle - \nu \langle \Delta u, \phi \rangle + \langle (u \cdot \nabla)u, \phi \rangle + \langle \nabla p, \phi \rangle = 0.$$

The pressure term $\langle \nabla p, \phi \rangle$ is then canceled by Corollary 2.14. We transform the viscous term:

$$-\nu \langle \Delta u, \phi \rangle = \nu \langle \nabla u, \nabla \phi \rangle$$

Since $\phi \in V$ does not depend on time, $\partial_t \phi = 0$, we integrate in time from 0 to τ and invoke the fundamental theorem of calculus:

$$(3.2) \quad \int_0^\tau \langle \partial_t u, \phi \rangle dt = \langle u(\tau), \phi \rangle - \langle u(0), \phi \rangle$$

We get that for any $\tau \geq 0$:

$$(3.3) \quad \langle u(\tau), \phi \rangle - \langle u(0), \phi \rangle + \nu \int_0^\tau \langle \nabla u, \nabla \phi \rangle dt + \int_0^\tau \langle (u \cdot \nabla)u, \phi \rangle dt = 0$$

(3.3) should be satisfied $\forall \phi \in V$. Since u was assumed smooth, $u(0)$ is well defined, and we write $u_0 = u(0)$ for the initial datum; for the weak solutions defined below no pointwise value at $t = 0$ is available a priori, so u_0 is instead prescribed as data. Every classical solution of Navier–Stokes equations satisfies (3.3) for all test functions; the converse is not true.

This makes the class of eligible weak solutions large and sometimes contradictory, therefore we attempt to narrow it by constraining the requirements.

We must establish a foundational result that lets us proceed onwards.

LEMMA 3.5. *For $u, v, w \in \dot{H}^1(\mathbb{T}^2)$, $|b(u, v, w)| \leq C\|u\|_{\dot{H}^1}\|v\|_{\dot{H}^1}\|w\|_{\dot{H}^1}$. Hence, b is continuous on $\dot{H}^1 \times \dot{H}^1 \times \dot{H}^1$.*

Proof. Hölder with exponents $(4, 2, 4)$ gives $|b(u, v, w)| \leq \|u\|_{\mathbb{L}^4}\|\nabla v\|_{\mathbb{L}^2}\|w\|_{\mathbb{L}^4}$. Applying Theorem 2.18 to u and w gives:

$$(3.4) \quad |b(u, v, w)| \leq C\|u\|_{\mathbb{L}^2}^{1/2}\|\nabla u\|_{\mathbb{L}^2}^{1/2}\|\nabla v\|_{\mathbb{L}^2}\|w\|_{\mathbb{L}^2}^{1/2}\|\nabla w\|_{\mathbb{L}^2}^{1/2} \leq C\|u\|_{\dot{H}^1}\|v\|_{\dot{H}^1}\|w\|_{\dot{H}^1}. \quad \square$$

LEMMA 3.6. *Let $u \in V$ and $v, w \in \dot{H}^1$. Then,*

$$\langle (u \cdot \nabla)v, w \rangle = -\langle (u \cdot \nabla)w, v \rangle$$

In particular:

$$\langle (u \cdot \nabla)v, v \rangle = 0.$$

Proof. For $u \in C_\sigma^\infty(\mathbb{T}^2)$ and $v, w \in C^1(\mathbb{T}^2)$, we find the $\mathbb{L}^2$ inner product using Einstein's summation convention:

$$\langle (u \cdot \nabla)v, w \rangle = \int u_j(\partial_j v_i)w_i = \int (\partial_j v_i)[u_j w_i]$$

Let $\Psi_j = v_i u_j w_i$, then we consider

$$\int_{\mathbb{T}^2} \partial_j \Psi_j dx.$$

On the periodic torus, the integral of the derivative vanishes:

$$\int_{\mathbb{T}^2} \partial_j \Psi_j dx = 0$$

Therefore, we get:

$$0 = \int \partial_j (v_i [u_j w_i])$$

Expand with a product rule:

$$\partial_j (v_i [u_j w_i]) = \partial_j v_i [u_j w_i] + v_i \partial_j [u_j w_i]$$

Returning the expression inside the integral:

$$\int (\partial_j v_i [u_j w_i] + v_i \partial_j [u_j w_i]) = 0$$

This yields:

$$\int (\partial_j v_i [u_j w_i]) = - \int (v_i \partial_j [u_j w_i])$$

Opening the derivative via product rule:

$$\partial_j [u_j w_i] = (\partial_j u_j) w_i + u_j (\partial_j w_i)$$

Returning the expression inside the integral:

$$\int (\partial_j v_i [u_j w_i]) = - \int v_i (\partial_j u_j) w_i - \int v_i u_j (\partial_j w_i)$$

The term $-\int v_i (\partial_j u_j) w_i$ contains $\nabla \cdot u$ summed over x_j , which is zero since $u \in V$. Therefore, only the second term is non-zero:

$$\int (\partial_j v_i [u_j w_i]) = - \int v_i u_j (\partial_j w_i)$$

Returning to the original expression:

$$\langle (u \cdot \nabla) v, w \rangle = - \langle (u \cdot \nabla) w, v \rangle$$

For $u \in V$ and $v, w \in \mathbb{H}^1$, we may use a density argument since the trilinear form:

$$b(u, v, w) = \langle (u \cdot \nabla) v, w \rangle$$

is continuous on $\mathbb{H}^1 \times \mathbb{H}^1 \times \mathbb{H}^1$ (Lemma 3.5).

Set $w = v$, and upon repeating the operations, we notice:

$$\langle (u \cdot \nabla) v, v \rangle = - \langle (u \cdot \nabla) v, v \rangle,$$

which gives $2 \langle (u \cdot \nabla) v, v \rangle = 0$. □

Assume that u is a classical solution of the Navier–Stokes equations; then by taking the $\mathbb{L}^2$ inner product with u we obtain:

$$\langle \partial_t u, u \rangle - \nu \langle \Delta u, u \rangle + \langle (u \cdot \nabla) u, u \rangle + \langle \nabla p, u \rangle = 0$$

By the chain rule, we have:

$$\langle \partial_t u, u \rangle = \frac{1}{2} \frac{d}{dt} \|u\|^2$$

The pressure term $\langle \nabla p, u \rangle$ vanishes in the computations because the spaces H and G are orthogonal (Corollary 2.14). Additionally, by Lemma 3.6, $\langle (u \cdot \nabla) u, u \rangle = 0$. Finally, the term $\nu \langle \Delta u, u \rangle$ is transformed:

$$-\nu \langle \Delta u, u \rangle = \nu \langle \nabla u, \nabla u \rangle = \nu \|\nabla u\|^2$$

Thus, the expression reduces to:

$$\frac{1}{2} \frac{d}{dt} \|u\|^2 + \nu \|\nabla u\|^2 = 0$$

Integrate the expression in time on the interval $[0, t]$:

$$\int_0^t \left(\frac{1}{2} \frac{d}{ds} \|u(s)\|^2 + \nu \|\nabla u(s)\|^2 \right) ds = 0$$

By fundamental theorem of calculus:

$$\frac{1}{2} \int_0^t \frac{d}{ds} \|u(s)\|^2 ds = \frac{1}{2} (\|u(t)\|^2 - \|u(0)\|^2)$$

Arranging the terms:

$$(3.5) \quad \frac{1}{2} \|u(t)\|^2 + \nu \int_0^t \|\nabla u(s)\|^2 ds = \frac{1}{2} \|u(0)\|^2$$

for any positive t . From this formulation, we naturally require that both quantities should be finite $\forall T > 0$:

$$\operatorname{ess\,sup}_{0 \leq t \leq T} \|u(t)\| < \infty \text{ and } \int_0^T \|\nabla u(s)\|^2 ds < \infty$$

Hence, we should look for weak solutions in the intersection of the spaces

$$L^\infty(0, T; H) \cap L^2(0, T; V).$$

We are ready to define the following.

DEFINITION 3.7 (Weak solution). *We say that a function u is a weak solution of the Navier–Stokes equations with initial condition $u_0 \in H$ if the following hold.*

(i) $u \in L^\infty(0, T; H) \cap L^2(0, T; V)$ for every $T > 0$;

(ii) u satisfies the identity

$$(3.6) \quad \langle u(\tau), \phi \rangle - \langle u_0, \phi \rangle + \nu \int_0^\tau \langle \nabla u, \nabla \phi \rangle dt + \int_0^\tau \langle (u \cdot \nabla) u, \phi \rangle dt = 0$$

for all test functions $\phi \in V$ and almost every $\tau > 0$.

The zero-mean condition requires no separate hypothesis: both H and V are defined through the homogeneous spaces of Definition 2.5, so any u satisfying (i) has zero mean for almost every $t > 0$.

4. Galerkin method. Our definition of the weak solution on a periodic torus $\mathbb{T}^2$ guarantees that at each time $t > 0$, the weak solution of Navier–Stokes equations $u(t)$ belongs to the infinite-dimensional space H spanned by the eigenfunctions of the Stokes operator (Definition 3.1). The Galerkin method allows us to construct a weak solution u as the limit of a sequence of approximate solutions u_n , that for each time $t > 0$ belong to the finite-dimensional space $P_n H$ spanned by the first n eigenfunctions:

$$P_n H := \operatorname{span}(a_1, a_2, \dots, a_n), \quad a_i \in \mathcal{N}$$

The basis $\mathcal{N}$ is given by Theorem 3.4; by P_n we denote the projection operator $P_n : \dot{\mathbb{L}}^2 \rightarrow H$ defined by (not to be confused with the Leray projector $\mathbb{P}$ of Definition 2.9):

$$(4.1) \quad P_n u = \sum_{i=1}^n \langle u, a_i \rangle a_i, \text{ where } a_i \in \mathcal{N}.$$

We usually consider the projection with domain $\dot{\mathbb{L}}^2$:

$$P_n : \dot{\mathbb{L}}^2 \rightarrow P_n H$$

Since on the torus $\mathbb{T}^2$ the functions a_i are smooth, we have that $P_n H \subset \mathbb{H}^k$ for every $k \in \mathbb{N}$. It is also straightforward that

$$\langle P_n u, v \rangle = \langle u, P_n v \rangle$$

for $u, v \in \dot{\mathbb{L}}^2$. We follow it up with a small lemma.

LEMMA 4.1. *If $u \in \mathbb{H}^s \cap H$, then $P_n u \rightarrow u$ as $n \rightarrow \infty$ in $\mathbb{H}^s(\mathbb{T}^2)$. Moreover,*

$$\|P_n u\|_{\mathbb{H}^s} \leq \|u\|_{\mathbb{H}^s}.$$

Proof. Each a_j is a single Fourier mode $m_{k(j)} \cos(k(j) \cdot x)$ or $m_{k(j)} \sin(k(j) \cdot x)$, so P_n is the orthogonal projection onto the span of the first n such modes, and expanding u in the basis $\mathcal{N}$ gives

$$\|u - P_n u\|_{\mathbb{H}^s}^2 = \sum_{j>n} (1 + \lambda_j)^s |\langle u, a_j \rangle|^2 \rightarrow 0$$

as $n \rightarrow \infty$, this being the tail of a convergent series — convergent precisely because $u \in \mathbb{H}^s$. The operator bound is then Bessel's inequality applied to the same expansion:

$$\|P_n u\|_{\mathbb{H}^s}^2 = \sum_{j \leq n} (1 + \lambda_j)^s |\langle u, a_j \rangle|^2 \leq \|u\|_{\mathbb{H}^s}^2 \quad \square$$

Now, we define the Galerkin equation, which can be thought of as the projection of Navier–Stokes equations onto the finite-dimensional space $P_n H$.

DEFINITION 4.2. *The n th-order Galerkin equation corresponding to Navier–Stokes equations is given by:*

$$\partial_t u_n + \nu A u_n + P_n[(u_n \cdot \nabla) u_n] = 0, \quad u_n(0) = P_n u_0$$

where A is the Stokes operator (Definition 3.1). We call u_n Galerkin approximations.

We test against each basis element a_j of $P_n H$, receiving n finite-dimensional ODEs:

$$(4.2) \quad \frac{d}{dt} \langle u_n, a_j \rangle + \nu \langle \nabla u_n, \nabla a_j \rangle + b(u_n, u_n, a_j) = 0, \quad j = 1, \dots, n,$$

with $u_n(0) = P_n u_0$. By construction $u_n(t) \in P_n H$ for every t , so it is determined by its n coordinates in the basis $\mathcal{N}$. Setting $c_j(t) = \langle u_n(t), a_j \rangle$ we may express u_n as:

$$u_n = \sum_{j=1}^n c_j(t) a_j$$

Note that this expansion involves only u_n ; the limit u has not yet been constructed, and Lemma 4.1 will be applied to the initial datum alone. Substituting into (4.2):

$$\frac{d}{dt} \left\langle \sum_{i=1}^n c_i a_i, a_j \right\rangle + \nu \left\langle \nabla \sum_{i=1}^n c_i a_i, \nabla a_j \right\rangle + b \left(\sum_{i=1}^n c_i a_i, \sum_{k=1}^n c_k a_k, a_j \right) = 0, \quad j = 1, \dots, n.$$

We address each expression one-by-one. First, the inner product is bilinear, so we may pull the time-dependent coefficients outside it; by orthonormality of $\mathcal{N}$ only the $i = j$ term survives:

$$\frac{d}{dt} \left\langle \sum_{i=1}^n c_i a_i, a_j \right\rangle = \frac{d}{dt} c_j = c'_j$$

Then, we perform the same in the viscous term:

$$\nu \left\langle \nabla \sum_{i=1}^n c_i a_i, \nabla a_j \right\rangle = \nu \sum_{i=1}^n c_i \langle \nabla a_i, \nabla a_j \rangle$$

Theorem 3.4 gives us $\langle \nabla a_i, \nabla a_j \rangle = \langle Aa_i, a_j \rangle$. We invoke the property:

$$\langle Aa_i, a_j \rangle = \lambda_i \delta_{ij}$$

where λ_i is the corresponding eigenvalue of a_i and δ_{ij} is the Kronecker delta. After all substitutions, we have:

$$\nu \sum_{i=1}^n c_i \langle \nabla a_i, \nabla a_j \rangle = \nu \lambda_j c_j$$

Finally, we perform analogous operations on the trilinear operator:

$$b \left(\sum_{i=1}^n c_i a_i, \sum_{k=1}^n c_k a_k, a_j \right) = \sum_{i=1}^n \sum_{k=1}^n c_i c_k b(a_i, a_k, a_j)$$

Operator $b(a_i, a_k, a_j)$ is not time-dependent by definition, therefore we let:

$$B_{ikj} := b(a_i, a_k, a_j)$$

Finally, we get the full expression:

$$c'_j + \nu \lambda_j c_j + \sum_{i=1}^n \sum_{k=1}^n B_{ikj} c_i c_k = 0 \text{ for } j = 1, \dots, n$$

The initial conditions of this system are found by projecting the initial data:

$$c_j(0) = \langle P_n u_0, a_j \rangle = \langle u_0, a_j \rangle, \quad u_0 = u(0)$$

This yields:

$$(4.3) \quad c'_j = -\nu \lambda_j c_j - \sum_{i=1}^n \sum_{k=1}^n B_{ikj} c_i c_k$$

Define the time-dependent vector of coefficients $c(t) \in \mathbb{R}^n$:

$$c(t) = \begin{bmatrix} c_1(t) \\ \vdots \\ c_n(t) \end{bmatrix}$$

Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$, where the j -th component of $F(c)$ encapsulates the spatial mechanics of Navier–Stokes equations so that:

$$F_j(c) = -\nu \lambda_j c_j - \sum_{i=1}^n \sum_{k=1}^n B_{ikj} c_i c_k$$

We have:

$$\frac{dc}{dt} = F(c), \quad c(0) = c_0$$

We are ready to formulate the theorem about the local existence of solutions.

THEOREM 4.3 (Local existence for the Galerkin system). *Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be defined as above. For any initial datum $c_0 \in \mathbb{R}^n$ there exist a maximal time $T_n \in (0, \infty]$, depending on n and c_0 , and a unique C^1 function $c : [0, T_n) \rightarrow \mathbb{R}^n$ solving $\frac{dc}{dt} = F(c)$, $c(0) = c_0$. If $T_n < \infty$ then $|c(t)| \rightarrow \infty$ as $t \rightarrow T_n^-$.*

Proof. Each F_j is a polynomial in the components of c , hence continuously differentiable and so locally Lipschitz on $\mathbb{R}^n$. The Picard–Lindelöf theorem then supplies a unique solution on a maximal interval of existence, which is open at the right endpoint, and the standard continuation criterion gives the blow-up alternative. Since F is continuous and c solves the equation, c is in fact C^1 rather than merely absolutely continuous [11]. $\square$

We return to the n th-order Galerkin equation and find its inner product with u_n :

$$\langle \partial_t u_n, u_n \rangle + \nu \langle Au_n, u_n \rangle + \langle P_n[(u_n \cdot \nabla)u_n], u_n \rangle = 0$$

By Lemma 3.6, term $\langle (u_n \cdot \nabla)u_n, u_n \rangle = 0$. We transform the time derivative via chain rule:

$$\langle \partial_t u_n, u_n \rangle = \frac{1}{2} \frac{d}{dt} \|u_n\|^2$$

Finally, we open the viscous term:

$$\nu \langle Au_n, u_n \rangle = \nu \langle -\mathbb{P}\Delta u_n, u_n \rangle = \nu \langle -\Delta u_n, u_n \rangle = \nu \langle \nabla u_n, \nabla u_n \rangle = \nu \|\nabla u_n\|^2$$

This evaluates to:

$$(4.4) \quad \frac{1}{2} \frac{d}{dt} \|u_n\|^2 + \nu \|\nabla u_n\|^2 = 0$$

As kinematic viscosity and the gradient norm squared are always non-negative, this implies:

$$(4.5) \quad \frac{d}{dt} \|u_n(t)\|_{\mathbb{L}^2}^2 \leq 0$$

Integrating in time on $[0, t]$ and invoking fundamental theorem of calculus:

$$\int_0^t \frac{d}{ds} \|u_n(s)\|_{\mathbb{L}^2}^2 ds = \|u_n(t)\|_{\mathbb{L}^2}^2 - \|u_n(0)\|_{\mathbb{L}^2}^2$$

This evaluates to:

$$(4.6) \quad \begin{aligned} \|u_n(t)\|_{\mathbb{L}^2}^2 - \|u_n(0)\|_{\mathbb{L}^2}^2 &\leq 0 \\ \|u_n(t)\|_{\mathbb{L}^2}^2 &\leq \|u_n(0)\|_{\mathbb{L}^2}^2 \leq \|u_0\|_{\mathbb{L}^2}^2 \end{aligned}$$

We formalize this result.

LEMMA 4.4 (Global existence for the Galerkin system). *The maximal time T_n of Theorem 4.3 is infinite, and for every $T > 0$*

$$u_n \in L^\infty(0, T; H), \quad \sup_{t \geq 0} \|u_n(t)\|_{\mathbb{L}^2} \leq \|u_0\|_{\mathbb{L}^2}.$$

Proof. By (4.6), $\|u_n(t)\|_{\mathbb{L}^2} \leq \|u_0\|_{\mathbb{L}^2}$ for every t in the interval of existence. Since $\mathcal{N}$ is orthonormal, $|c(t)|^2 = \sum_{j \leq n} c_j(t)^2 = \|u_n(t)\|_{\mathbb{L}^2}^2$, so the coefficient vector stays in a fixed ball of $\mathbb{R}^n$. The blow-up alternative of Theorem 4.3 therefore rules out $T_n < \infty$, and $T_n = \infty$.

The bound holds for every t , not merely almost every t , so the supremum above is genuine and in particular

$$\operatorname{ess\,sup}_{0 \leq t \leq T} \|u_n(t)\|_{\mathbb{L}^2} \leq \|u_0\|_{\mathbb{L}^2} < \infty$$

for every $T > 0$, which is the assertion $u_n \in L^\infty(0, T; H)$. $\square$

Returning to (4.4), we integrate the equation in time on the interval $[0, T]$ and invoke the fundamental theorem of calculus, thus expressing the gradient bound:

$$\begin{aligned} \frac{1}{2} \|u_n(T)\|^2 - \frac{1}{2} \|u_n(0)\|^2 + \nu \int_0^T \|\nabla u_n\|^2 dt &= 0 \\ \nu \int_0^T \|\nabla u_n\|^2 dt &= \frac{1}{2} \|u_n(0)\|^2 - \frac{1}{2} \|u_n(T)\|^2 \end{aligned}$$

By (4.6) the right-hand side is non-negative and at most $\frac{1}{2} \|u_0\|^2$, since $\|u_n(0)\| \leq \|u_0\|$ and $\|u_n(T)\|^2 \geq 0$. Discarding the subtracted term gives:

$$(4.7) \quad \begin{aligned} \nu \int_0^T \|\nabla u_n\|^2 dt &\leq \frac{1}{2} \|u_0\|^2 \\ \int_0^T \|\nabla u_n\|_{\mathbb{L}^2}^2 dt &\leq \frac{1}{2\nu} \|u_0\|_{\mathbb{L}^2}^2 \end{aligned}$$

We formalize this result.

LEMMA 4.5 (Bound on the velocity gradient). *For every $T > 0$ the Galerkin approximations satisfy*

$$u_n \in L^\infty(0, T; H) \cap L^2(0, T; V),$$

with bounds depending only on $\|u_0\|_{\mathbb{L}^2}$, ν and T , and not on n .

Proof. (4.7) bounds $\int_0^T \|\nabla u_n\|_{\mathbb{L}^2}^2 dt$ by $\frac{1}{2\nu} \|u_0\|_{\mathbb{L}^2}^2$, and $\|\nabla \cdot\|_{\mathbb{L}^2}$ is the norm of V (Definition 2.15); this is the $L^2(0, T; V)$ bound. Lemma 4.4 supplies the $L^\infty(0, T; H)$ bound. Neither depends on n , which is what the compactness argument of section 5 requires. $\square$

5. Existence of weak solutions. We establish the following result that proves to be foundational.

LEMMA 5.1. *Suppose that (u_n) is bounded in $L^2(0, T; \mathbb{L}^2)$ for every $T > 0$. Then, there exists a subsequence of (u_n) that is weakly convergent in $L^2(0, T; \mathbb{L}^2)$ for every $T > 0$.*

Proof. The space $L^2(0, T; \mathbb{L}^2)$ is a separable Hilbert space, so it contains a countable dense subset

$$\{\phi_j\}_{j \in \mathbb{N}} = \{\phi_1, \phi_2, \phi_3, \dots\}$$

which is infinite; the extraction below accordingly runs through infinitely many stages. We test the bounded Galerkin sequence (u_n) against the first test function of the dense subset, which is ϕ_1 , yielding a sequence given as:

$$\langle u_n, \phi_1 \rangle$$

This sequence of scalars is bounded, since $|\langle u_n, \phi_1 \rangle| \leq \|u_n\| \|\phi_1\|$ and (u_n) is bounded by hypothesis. By Theorem 2.19 applied to those scalars, we may extract a subsequence of (u_n) , written $(u_n^{(1)})_{n \in \mathbb{N}}$, along which $\langle u_n^{(1)}, \phi_1 \rangle$ converges.

We now test $(u_n^{(1)})$ against the second test function ϕ_2 , obtaining the bounded scalar sequence $\langle u_n^{(1)}, \phi_2 \rangle$. Applying Theorem 2.19 again, we extract a further subsequence $(u_n^{(2)})_{n \in \mathbb{N}}$ along which $\langle u_n^{(2)}, \phi_2 \rangle$ also converges. Iterating over $j \in \mathbb{N}$ produces a nested family of infinite subsequences

$$(5.1) \quad \begin{aligned} & (u_n^{(1)})_{n \in \mathbb{N}} \supset (u_n^{(2)})_{n \in \mathbb{N}} \supset (u_n^{(3)})_{n \in \mathbb{N}} \supset \dots \\ & \text{with } \langle u_n^{(j)}, \phi_i \rangle \text{ convergent as } n \rightarrow \infty \text{ for every } i \leq j. \end{aligned}$$

Each row of (5.1) is an infinite sequence, and each is a subsequence of the one above it. We now form the diagonal sequence by taking the m th term of the m th row:

$$(v_m) = (u_1^{(1)}, u_2^{(2)}, u_3^{(3)}, u_4^{(4)}, \dots)$$

For each fixed j , the tail $(v_m)_{m \geq j}$ is a subsequence of $(u_n^{(j)})_n$, and a convergent sequence has all its subsequences converging to the same limit. Hence $\langle v_m, \phi_j \rangle$ converges for every $j \in \mathbb{N}$. Since $\{\phi_j\}$ is dense and (v_m) is bounded, an $\varepsilon/3$ argument extends the convergence of $\langle v_m, \phi \rangle$ to every $\phi \in L^2(0, T; \mathbb{L}^2)$, and the Riesz representation theorem identifies the limit functional with some u in that space. Thus $v_m \rightharpoonup u$. $\square$

We return to the n th-order Galerkin equation and take the inner product with the test function $\phi \in V$. We use the orthogonal projection P_n for the test function:

$$\begin{aligned} \langle \partial_t u_n, \phi \rangle + \nu \langle Au_n, P_n \phi \rangle + \langle (u_n \cdot \nabla) u_n, P_n \phi \rangle &= 0 \\ \langle \partial_t u_n, \phi \rangle &= -\nu \langle \nabla u_n, P_n \phi \rangle - b(u_n, u_n, P_n \phi) \end{aligned}$$

to invoke the bound:

$$\|P_n \phi\|_V \leq \|\phi\|_V \leq 1$$

Using Cauchy-Schwarz inequality and Definition 2.15, we transform the viscous term:

$$-\nu \langle \nabla u_n, P_n \phi \rangle \leq \nu \|\nabla u_n\|_{\mathbb{L}^2} \|\nabla P_n \phi\|_{\mathbb{L}^2} = \nu \|\nabla u_n\|_{\mathbb{L}^2} \|\phi\|_V$$

We bound the trilinear form and use integration by parts to shift the ∇ operator:

$$\begin{aligned} b(u_n, u_n, P_n \phi) &= -b(u_n, P_n \phi, u_n) \\ b(u_n, u_n, P_n \phi) &= - \int_{\mathbb{T}^2} (u_n \cdot \nabla) P_n \phi \cdot u_n dx \end{aligned}$$

We take Hölder exponents of $(4, 4, 2)$ since $\frac{1}{4} + \frac{1}{4} + \frac{1}{2} = 1$:

$$\begin{aligned} \int_{\mathbb{T}^2} (u_n \cdot \nabla) P_n \phi \cdot u_n dx &\leq \int_{\mathbb{T}^2} |u_n| |u_n| |\nabla P_n \phi| dx \\ \int_{\mathbb{T}^2} |u_n| |u_n| |\nabla P_n \phi| dx &\leq \|u_n\|_{\mathbb{L}^4} \|u_n\|_{\mathbb{L}^4} \|\nabla P_n \phi\|_{\mathbb{L}^2} = \|u_n\|_{\mathbb{L}^4}^2 \|\nabla P_n \phi\|_{\mathbb{L}^2} \end{aligned}$$

Using Theorem 2.18:

$$\|u_n\|_{\mathbb{L}^4}^2 \|\nabla P_n \phi\|_{\mathbb{L}^2} \leq C \|u_n\|_{\mathbb{L}^2} \|\nabla u_n\|_{\mathbb{L}^2} \|\nabla P_n \phi\|_{\mathbb{L}^2} = C \|u_n\|_{\mathbb{L}^2} \|\nabla u_n\|_{\mathbb{L}^2} \|\phi\|_V$$

Connecting the expressions together, for every ϕ with $\|\phi\|_V \leq 1$:

$$|\langle \partial_t u_n, \phi \rangle| \leq \left(\nu + C \|u_n\|_{\mathbb{L}^2} \right) \|\nabla u_n\|_{\mathbb{L}^2} \|\phi\|_V \leq \left(\nu + C \|u_n\|_{\mathbb{L}^2} \right) \|\nabla u_n\|_{\mathbb{L}^2}$$

The right-hand side no longer involves ϕ , so we may take the supremum over all such ϕ and invoke the definition of the dual norm from Definition 2.15:

$$\|\partial_t u_n\|_{V'} = \sup_{\|\phi\|_V \leq 1} |\langle \partial_t u_n, \phi \rangle| \leq \left(\nu + C \|u_n\|_{\mathbb{L}^2} \right) \|\nabla u_n\|_{\mathbb{L}^2}$$

By Lemma 4.4, $\|u_n(t)\|_{\mathbb{L}^2} \leq \|u_0\|_{\mathbb{L}^2}$ for every t , so the bracket is bounded by the constant $K = \nu + C \|u_0\|_{\mathbb{L}^2}$, uniformly in t and in n . This is what allows it to leave the time integral. Squaring and integrating on $[0, T]$:

$$\int_0^T \|\partial_t u_n\|_{V'}^2 dt \leq K^2 \int_0^T \|\nabla u_n\|_{\mathbb{L}^2}^2 dt \leq \frac{K^2}{2\nu} \|u_0\|_{\mathbb{L}^2}^2$$

using (4.7) from Lemma 4.5 in the last step. The right-hand side is a constant independent of n .

This gives us a bound on the time derivative:

$$(5.2) \quad \partial_t u_n \in L^2(0, T; V')$$

We refer to the known theorem that lets us recover strong convergence of the sequences in L^2 space.

THEOREM 5.2 (Aubin–Lions, simple version [1, 11]). *If the sequence (u_n) is uniformly bounded in $L^q(0, T; V)$, $q > 1$, and the partial derivatives $\partial_t u_n$ are uniformly bounded in $L^p(0, T; V')$, $p > 1$, then there exist $u \in L^q(0, T; H)$ and a subsequence of (u_n) that converges strongly to u in $L^q(0, T; H)$.*

As both hypotheses of the theorem were fulfilled, the theorem is instantly applicable. Thus, we are ready to state the first key result of this paper.

THEOREM 5.3 (Hopf [7, 9]). *For every $u_0 \in H(\mathbb{T}^2)$, there exists at least one global-in-time weak solution of the Navier–Stokes equations satisfying the initial condition u_0 .*

Proof. By Theorem 5.2, Theorem 2.20 and Corollary 2.21, we may pass to a subsequence of (u_n) , not relabelled, such that the following hold:

$$(5.3) \quad \begin{aligned} u_n &\rightarrow u \text{ strongly in } L^2(0, T; H(\mathbb{T}^2)) \\ u_n &\overset{*}{\rightharpoonup} u \text{ weakly-}^* \text{ in } L^\infty(0, T; H(\mathbb{T}^2)) \\ \nabla u_n &\rightharpoonup \nabla u \text{ weakly in } L^2(0, T; \mathbb{L}^2(\mathbb{T}^2)) \end{aligned}$$

Passing to a further subsequence if necessary, strong convergence in $L^2(0, T; H)$ gives convergence pointwise in time along that subsequence:

$$(5.4) \quad u_n(t) \rightarrow u(t) \text{ in } \mathbb{L}^2(\mathbb{T}^2) \text{ for almost every } t \in [0, \infty).$$

From Definition 3.7, we know that the weak solution of Navier–Stokes equations must satisfy, for all $\phi \in V$:

$$\langle u(\tau), \phi \rangle - \langle u_0, \phi \rangle + \nu \int_0^\tau \langle \nabla u, \nabla \phi \rangle dt + \int_0^\tau \langle (u \cdot \nabla)u, \phi \rangle dt = 0$$

Fix a test function $\phi \in C_c^\infty(\mathbb{T}^2)$. Such ϕ are dense in V , and taking them smooth makes the $\mathbb{L}^\infty$ bound used below available; the identity for general $\phi \in V$ follows at the end by density and the continuity of every term in $\|\phi\|_V$.

The n th-order Galerkin equation holds after testing against elements of $P_n H$, so pairing it with ϕ places P_n on the test function:

$$\begin{aligned} \langle \partial_t u_n, \phi \rangle + \nu \langle A u_n, P_n \phi \rangle + \langle (u_n \cdot \nabla)u_n, P_n \phi \rangle &= 0 \\ \langle \partial_t u_n, \phi \rangle + \nu \langle \nabla u_n, \nabla P_n \phi \rangle + \langle (u_n \cdot \nabla)u_n, P_n \phi \rangle &= 0 \end{aligned}$$

using $\langle \partial_t u_n, \phi \rangle = \langle \partial_t u_n, P_n \phi \rangle$, valid because $\partial_t u_n \in P_n H$. By Lemma 4.1, $P_n \phi \rightarrow \phi$ in $\mathbb{H}^1$, so replacing $P_n \phi$ by ϕ in the limit is legitimate; we do so silently below.

Integrating in time on $[0, \tau]$, to match Definition 3.7's formulation:

$$\int_0^\tau \langle \partial_t u_n, \phi \rangle dt + \nu \int_0^\tau \langle \nabla u_n, \nabla P_n \phi \rangle dt + \int_0^\tau \langle (u_n \cdot \nabla)u_n, P_n \phi \rangle dt = 0$$

By fundamental theorem of calculus, analogous to (3.2):

$$\int_0^\tau \langle \partial_t u_n, \phi \rangle dt = \langle u_n(\tau), \phi \rangle - \langle u_n(0), \phi \rangle$$

Using the formulation given by (5.4), $\langle u_n(\tau), \phi \rangle \rightarrow \langle u(\tau), \phi \rangle$ for almost every τ . The initial term is handled separately: $u_n(0) = P_n u_0 \rightarrow u_0$ in $\mathbb{L}^2$ by Lemma 4.1, so $\langle u_n(0), \phi \rangle \rightarrow \langle u_0, \phi \rangle$, and it is in this sense that the datum is attained. Writing $u_0 = u(0)$, we get:

$$\langle u(\tau), \phi \rangle - \langle u(0), \phi \rangle + \nu \int_0^\tau \langle \nabla u_n, \nabla P_n \phi \rangle dt + \int_0^\tau \langle (u_n \cdot \nabla)u_n, P_n \phi \rangle dt = 0$$

From (5.3), we know that

$$\langle \nabla u_n, \nabla \phi \rangle \rightarrow \langle \nabla u, \nabla \phi \rangle$$

Finally, we only have to prove that:

$$\int_0^\tau \langle (u_n \cdot \nabla)u_n, \phi \rangle dt \rightarrow \int_0^\tau \langle (u \cdot \nabla)u, \phi \rangle dt$$

We note that, by adding and subtracting:

$$\begin{aligned} (5.5) \quad (u_n \cdot \nabla)u_n - (u \cdot \nabla)u &= (u_n \cdot \nabla)u_n + (u \cdot \nabla)u_n - (u \cdot \nabla)u_n - (u \cdot \nabla)u \\ &= ((u_n - u) \cdot \nabla)u_n + (u \cdot \nabla)(u_n - u) \end{aligned}$$

Therefore we need to show:

$$\int_0^\tau \langle ((u_n - u) \cdot \nabla) u_n, \phi \rangle dt \rightarrow 0 \text{ and } \int_0^\tau \langle (u \cdot \nabla)(u_n - u), \phi \rangle dt \rightarrow 0$$

For the first, put $C_\phi = \|\phi\|_{\mathbb{L}^\infty}$, which is finite precisely because ϕ was taken smooth — note that $\dot{\mathbb{H}}^1(\mathbb{T}^2)$ does not embed in $\mathbb{L}^\infty$ in two dimensions, so this step is unavailable for general $\phi \in V$. Then by Cauchy-Schwarz in x :

$$\left| \int_0^\tau \langle ((u_n - u) \cdot \nabla) u_n, \phi \rangle dt \right| \leq C_\phi \int_0^\tau \|u_n - u\| \|\nabla u_n\| dt$$

By Hölder inequality with exponents $(2, 2)$:

$$C_\phi \int_0^\tau \|u_n - u\| \|\nabla u_n\| dt \leq C_\phi \left(\int_0^\tau \|u_n - u\|^2 dt \right)^{1/2} \left(\int_0^\tau \|\nabla u_n\|^2 dt \right)^{1/2} \rightarrow 0$$

For the second, integrate by parts using Lemma 3.6 to move the derivative off $u_n - u$:

$$\int_0^\tau \langle (u \cdot \nabla)(u_n - u), \phi \rangle dt = - \int_0^\tau \langle (u \cdot \nabla)\phi, u_n - u \rangle dt.$$

Here $(u \cdot \nabla)\phi \in L^2(0, T; \mathbb{L}^2)$, since $\nabla\phi$ is bounded and $u \in L^\infty(0, T; H)$, so this pairs against the strong convergence $u_n \rightarrow u$ in $L^2(0, T; H)$ and

$$\int_0^\tau \langle (u \cdot \nabla)(u_n - u), \phi \rangle dt \rightarrow 0.$$

Thus, it is clear that

$$\int_0^\tau \langle (u_n \cdot \nabla) u_n, \phi \rangle dt \rightarrow \int_0^\tau \langle (u \cdot \nabla) u, \phi \rangle dt$$

Passing to the limit in every term therefore gives, for almost every $\tau > 0$ and every $\phi \in C_\sigma^\infty(\mathbb{T}^2)$,

$$\langle u(\tau), \phi \rangle - \langle u_0, \phi \rangle + \nu \int_0^\tau \langle \nabla u, \nabla \phi \rangle dt + \int_0^\tau \langle (u \cdot \nabla) u, \phi \rangle dt = 0,$$

and by density this extends to all $\phi \in V$. This is part (ii) of Definition 3.7.

It remains to check part (i), and here the conclusion must be drawn about the limit rather than about the approximations. The bounds of Lemma 4.5 are uniform in n , so by Corollary 2.21 and Theorem 2.20 the limit inherits them: weak lower semicontinuity of the norm gives

$$\|u\|_{L^\infty(0, T; H)} \leq \liminf_{n \rightarrow \infty} \|u_n\|_{L^\infty(0, T; H)} \leq \|u_0\|_{\mathbb{L}^2},$$

and likewise $\int_0^T \|\nabla u\|^2 dt \leq \liminf_{n \rightarrow \infty} \int_0^T \|\nabla u_n\|^2 dt \leq \frac{1}{2\nu} \|u_0\|_{\mathbb{L}^2}^2$. Hence

$$u \in L^\infty(0, T; H) \cap L^2(0, T; V) \text{ for all } T > 0.$$

Both parts of Definition 3.7 hold, and since $T > 0$ was arbitrary the weak solution u is global in time. $\square$

We know that the classical solutions of the Navier–Stokes equations satisfy the energy equality:

$$(5.6) \quad \frac{1}{2}\|u(t)\|^2 + \nu \int_0^t \|\nabla u\|^2 ds = \frac{1}{2}\|u_0\|^2$$

Now we define the Leray–Hopf weak solutions as the solutions which satisfy the strong energy inequality, which is similar to (5.6).

DEFINITION 5.4 (Leray–Hopf weak solutions). *We say that u is a Leray–Hopf weak solution of the Navier–Stokes equations if u is a weak solution of the Navier–Stokes equations that satisfies the strong energy inequality:*

$$\frac{1}{2}\|u(t)\|^2 + \nu \int_s^t \|\nabla u\|^2 d\sigma \leq \frac{1}{2}\|u(s)\|^2 \text{ for all } t > s$$

for $s = 0$ and almost all times $s \in (0, \infty)$.

Now we establish that the solutions constructed in accordance to Theorem 5.3 satisfy Definition 5.4.

THEOREM 5.5. *The weak solution u of Navier–Stokes equations constructed by Theorem 5.3 satisfies the strong energy inequality:*

$$(5.7) \quad \frac{1}{2}\|u(t)\|^2 + \nu \int_s^t \|\nabla u\|^2 ds \leq \frac{1}{2}\|u(s)\|^2 \text{ for all } t > s$$

for $s = 0$ and almost all times $s \in (0, \infty)$. In particular, u satisfies

$$(5.8) \quad \frac{1}{2}\|u(t)\|^2 + \nu \int_0^t \|\nabla u\|^2 ds \leq \frac{1}{2}\|u(0)\|^2 \text{ for all } t > 0$$

Proof. Taking the inner product of the Galerkin equation with its solution u_n :

$$\langle \partial_t u_n, u_n \rangle + \nu \langle Au_n, u_n \rangle + \langle (u_n \cdot \nabla)u_n, u_n \rangle = 0$$

By Lemma 3.6, $\langle (u_n \cdot \nabla)u_n, u_n \rangle = 0$. We transform the first two terms:

$$\langle \partial_t u_n, u_n \rangle = \frac{1}{2} \frac{d}{dt} \|u_n\|^2$$

Similarly:

$$\nu \langle Au_n, u_n \rangle = \nu \langle \nabla u_n, \nabla u_n \rangle = \nu \|\nabla u_n\|^2$$

Thus, we get:

$$\frac{1}{2} \frac{d}{dt} \|u_n\|^2 + \nu \|\nabla u_n\|^2 = 0$$

We integrate in time on $[s, t]$ and invoke the fundamental theorem of calculus:

$$\frac{1}{2}\|u_n(t)\|^2 + \nu \int_s^t \|\nabla u_n\|^2 ds = \frac{1}{2}\|u_n(s)\|^2$$

We know that $\nabla u_n \rightharpoonup \nabla u$ in $L^2(s, t; \mathbb{L}^2)$, hence we can invoke weak lower semicontinuity of the norm: the norm of a weak limit is no larger than the limit inferior of the norms along the sequence,

$$x_n \rightharpoonup x \text{ in a Hilbert space } \mathcal{H} \implies \|x\|_{\mathcal{H}} \leq \liminf_{n \rightarrow \infty} \|x_n\|_{\mathcal{H}}.$$

Applying $\liminf$ to both sides of the equation:

$$\begin{aligned} \liminf_{n \rightarrow \infty} \left(\frac{1}{2} \|u_n(t)\|^2 + \nu \int_s^t \|\nabla u_n\|^2 ds \right) &= \liminf_{n \rightarrow \infty} \left(\frac{1}{2} \|u_n(s)\|^2 \right) \\ \liminf_{n \rightarrow \infty} \left(\frac{1}{2} \|u_n(t)\|^2 \right) + \nu \liminf_{n \rightarrow \infty} \left(\int_s^t \|\nabla u_n\|^2 ds \right) &\leq \frac{1}{2} \|u(s)\|^2 \\ \frac{1}{2} \|u(t)\|^2 + \nu \int_s^t \|\nabla u\|^2 ds &\leq \frac{1}{2} \|u(s)\|^2 \end{aligned}$$

This is valid for almost all s and almost all $t > s$. To prove for all $t > s$, we let sequence $t' \rightarrow t$ such that:

$$\frac{1}{2} \|u(t')\|^2 + \nu \int_s^{t'} \|\nabla u\|^2 ds \leq \frac{1}{2} \|u(s)\|^2$$

As u is L^2 -weakly continuous, we have:

$$\|u(t)\|^2 \leq \liminf_{t' \rightarrow t} \|u(t')\|^2$$

and

$$\int_s^{t'} \|\nabla u\|^2 ds \rightarrow \int_s^t \|\nabla u\|^2 ds$$

Thus:

$$\frac{1}{2} \|u(t)\|^2 + \nu \int_s^t \|\nabla u\|^2 ds \leq \liminf_{t' \rightarrow t} \left(\frac{1}{2} \|u(t')\|^2 + \nu \int_s^{t'} \|\nabla u\|^2 ds \right) \leq \frac{1}{2} \|u(s)\|^2. \quad \square$$

COROLLARY 5.6. *For the weak solutions constructed in Theorem 5.3 we have*

$$u(t) \rightarrow u(0) \text{ strongly in } \mathbb{L}^2(\mathbb{T}^2) \text{ as } t \rightarrow 0^+$$

Proof. From (5.7) with $s = 0$, multiplied by 2,

$$\|u(t)\|^2 + 2\nu \int_0^t \|\nabla u\|^2 ds \leq \|u(0)\|^2$$

Since the viscous term is non-negative, $\|u(t)\|^2 \leq \|u(0)\|^2$ for every $t > 0$, which implies

$$\limsup_{t \rightarrow 0^+} \|u(t)\| \leq \|u(0)\|$$

On the other hand, u is weakly continuous into $\mathbb{L}^2$, so $u(t) \rightharpoonup u(0)$ as $t \rightarrow 0^+$, and weak lower semicontinuity of the norm gives

$$\|u(0)\| \leq \liminf_{t \rightarrow 0^+} \|u(t)\|$$

The two displays together force $\|u(t)\| \rightarrow \|u(0)\|$. Thus we have both

$$u(t) \rightharpoonup u(0) \text{ in } \mathbb{L}^2(\mathbb{T}^2) \text{ and } \|u(t)\| \rightarrow \|u(0)\|,$$

which are convergence of the vectors and of their norms respectively — not, as the shorthand might suggest, weak convergence of the norms themselves. By Lemma 2.22, this yields

□

$$u(t) \rightarrow u(0) \text{ strongly in } \mathbb{L}^2(\mathbb{T}^2) \text{ as } t \rightarrow 0^+$$

6. Existence and properties of strong solutions. We begin this section by defining the notion of a strong solution from the standpoint of a weak solution.

DEFINITION 6.1 (Strong solution of Navier–Stokes). *We say that u is a strong solution of the Navier–Stokes equations on the time interval $[0, T]$ if u is a weak solution of Navier–Stokes equations and in addition:*

$$u \in L^\infty(0, T; \mathbb{H}^1) \cap L^2(0, T; \mathbb{H}^2)$$

We establish a desirable property of the strong solutions of the Navier–Stokes equations.

LEMMA 6.2. *Let $u \in L^2(0, T; \mathbb{H}^2)$ and $\partial_t u \in L^2(0, T; \mathbb{L}^2)$. Then,*

$$u \in C([0, T]; \mathbb{H}^1)$$

Proof. We shift the Gelfand triple by designating $\mathbb{H}^1$ as the pivot space, equipped with the inner product

$$\langle u, v \rangle_{\mathbb{H}^1} = \langle \nabla u, \nabla v \rangle_{\mathbb{L}^2} = \langle Au, v \rangle_{\mathbb{L}^2}.$$

With respect to this pivot space, the dual of $\mathbb{H}^2$ is isomorphic to $\mathbb{L}^2$, yielding a shifted Gelfand triple:

$$\mathbb{H}^2 \subset \mathbb{H}^1 \subset \mathbb{L}^2$$

Given our hypotheses $u \in L^2(0, T; \mathbb{H}^2)$ and $\partial_t u \in L^2(0, T; \mathbb{L}^2)$, all conditions of Theorem 2.17 are fulfilled. We immediately obtain $u \in C([0, T]; \mathbb{H}^1)$ and the absolutely continuous identity:

□

$$\frac{d}{dt} \|u\|_{\mathbb{H}^1}^2 = 2\langle \partial_t u, u \rangle_{\mathbb{L}^2, \mathbb{H}^2} = 2\langle \partial_t u, Au \rangle_{\mathbb{L}^2}.$$

We now consider more general properties of the strong solutions of Navier–Stokes equations.

LEMMA 6.3. *Suppose that u is a strong solution of the Navier–Stokes equations on $[0, T]$. Then*

$$\Delta u, \partial_t u, \text{ and } (u \cdot \nabla)u$$

are all elements of $L^2(0, T; \mathbb{L}^2)$.

Proof. From Definition 6.1, we immediately obtain:

$$\Delta u \in L^2(0, T; \mathbb{L}^2)$$

To prove $(u \cdot \nabla)u \in L^2(0, T; \mathbb{L}^2)$, we verify its square integrability:

$$\begin{aligned} \int_0^T \|(u(s) \cdot \nabla)u(s)\|^2 ds &\leq \int_0^T \int_{\mathbb{T}^2} |u(x, s)|^2 |\nabla u(x, s)|^2 dx ds \\ &\leq \int_0^T \|u(s)\|_{\mathbb{L}^\infty}^2 \int_{\mathbb{T}^2} |\nabla u(x, s)|^2 dx ds \\ &\leq \|u\|_{L^\infty(0, T; \mathbb{H}^1)}^2 \int_0^T \|u(s)\|_{\mathbb{L}^\infty}^2 ds \end{aligned}$$

By Corollary 2.23, $\mathbb{H}^2(\mathbb{T}^2) = [W^{2,2}(\mathbb{T}^2)]^2 \subset \mathbb{L}^\infty(\mathbb{T}^2)$, thus:

$$\begin{aligned} \|u\|_{L^\infty(0, T; \mathbb{H}^1)}^2 \int_0^T \|u(s)\|_{\mathbb{L}^\infty}^2 ds &\leq \|u\|_{L^\infty(0, T; \mathbb{H}^1)}^2 \int_0^T \|u(s)\|_{\mathbb{H}^2}^2 ds \\ &\leq \|u\|_{L^\infty(0, T; \mathbb{H}^1)}^2 \|u\|_{L^2(0, T; \mathbb{H}^2)}^2 \end{aligned}$$

Which satisfies Definition 6.1. Finally, we show that $\partial_t u$ is square-integrable. From Definition 3.7, we substitute $\tau = T$, thus getting:

$$\begin{aligned} \langle u(T), \phi \rangle &= \langle u(0), \phi \rangle - \nu \int_0^T \langle \nabla u, \nabla \phi \rangle dt - \int_0^T \langle (u \cdot \nabla)u, \phi \rangle dt \\ &= \langle u(0), \phi \rangle + \nu \int_0^T \langle \Delta u, \phi \rangle dt - \int_0^T \langle (u \cdot \nabla)u, \phi \rangle dt \end{aligned}$$

for all $T > 0$ and all $\phi \in V$. Differentiating with respect to time, we get:

$$\frac{d}{dt} \langle u(t), \phi \rangle = \langle \nu \Delta u - (u \cdot \nabla)u, \phi \rangle$$

Now let $w \in \mathbb{L}^2$ such that its Helmholtz–Leray decomposition is $w = \phi + \nabla v$. Applying the Leray projector $\mathbb{P}$ for both sides, we get that for almost all t :

$$\frac{d}{dt} \langle \mathbb{P}u(t), \mathbb{P}w \rangle = \langle g, \mathbb{P}w \rangle = \langle g, \phi \rangle$$

with $g = \mathbb{P}[\nu \Delta u - (u \cdot \nabla)u]$. Since for a strong solution the expression $\nu \Delta u - (u \cdot \nabla)u$ belongs to $L^2(0, T; \mathbb{L}^2)$, we conclude that $g \in L^2(0, T; H)$ and by Lemma 2.2, $\partial_t u \in L^2(0, T; H) \subset L^2(0, T; \mathbb{L}^2)$. $\square$

The next lemma shows that for almost all times the Leray projection of the function $\partial_t u - \nu \Delta u + (u \cdot \nabla)u$ is equal to zero.

LEMMA 6.4. *Suppose that u is a strong solution of the Navier–Stokes equations on $[0, T]$. Then*

$$\int_0^T \langle \partial_t u - \nu \Delta u + (u \cdot \nabla)u, w \rangle dt = 0$$

for all $w \in L^2(0, T; H)$.

Proof. By taking the n th-order Galerkin equation, passing a limit $n \rightarrow \infty$ (since $u_n \rightarrow u$ strongly, by Theorem 5.2 and (5.3)) and testing against $\phi \in V$, we get:

$$\langle \partial_t u(t) + \nu Au(t) + (u \cdot \nabla)u, \phi \rangle = 0$$

valid on $[0, T]$. For almost every t , $\partial_t u(t) \in \mathbb{L}^2$, $(u \cdot \nabla)u \in \mathbb{L}^2$, $Au \in \mathbb{L}^2$ by Lemma 6.3. The residual vector $R(t) = \partial_t u(t) + \nu Au(t) + (u \cdot \nabla)u$ therefore exists in $\mathbb{L}^2$. Moreover, the test space V is dense in H : given $u \in H$, the truncations $P_n u$ lie in $P_n H \subset V$ and converge to u in $\mathbb{L}^2$ by Lemma 4.1. (Mere inclusion $\mathbb{H}^1 \subset \mathbb{L}^2$ would not suffice.) The inner product is continuous in Hilbert spaces, and the only vector orthogonal to the dense subset of a space is the zero vector itself. Thus,

$$R(t) = 0 \implies \partial_t u(t) + \nu Au(t) + (u \cdot \nabla)u = 0$$

Transforming the Stokes operator into Laplacian and taking the integral on $[0, T]$:

$$\int_0^T \langle \partial_t u(t) - \nu \Delta u(t) + (u \cdot \nabla)u, w \rangle dt = \int_0^T \langle 0, w \rangle dt = 0.$$

for all $w \in L^2(0, T; H)$. $\square$

We prove that the strong solution of Navier–Stokes equations does not only satisfy an energy inequality (as is the case for Leray–Hopf weak solutions), but also an energy equality (as is the case for the classical solutions).

THEOREM 6.5. *If u is a strong solution of the Navier–Stokes equations on $[0, T]$, then it satisfies the energy equality*

$$(6.1) \quad \frac{1}{2} \|u(t_1)\|^2 + \nu \int_{t_0}^{t_1} \|\nabla u\|^2 ds = \frac{1}{2} \|u(t_0)\|^2$$

for all $t_0, t_1 \in [0, T]$, $t_0 < t_1$.

Proof. Let u be the strong solution of Navier–Stokes equations on $[0, T]$. Then, we take the inner product with the solution, valid on $[0, T]$:

$$\langle \partial_t u, u \rangle + \nu \langle Au, u \rangle + \langle (u \cdot \nabla)u, u \rangle = 0$$

Because $u \in C([0, T]; \mathbb{H}^1)$ and is absolutely continuous by Theorem 2.17, $\langle \partial_t u, u \rangle = \frac{1}{2} \frac{d}{dt} \|u\|^2$. Similarly, $\nu \langle Au, u \rangle = \nu \|\nabla u\|^2$. By Lemma 3.6, $\langle (u \cdot \nabla)u, u \rangle = 0$. Integrating in time on $[t_0, t_1]$ such that $0 \leq t_0 < t_1 \leq T$:

$$\int_{t_0}^{t_1} \frac{1}{2} \frac{d}{ds} \|u(s)\|^2 ds + \int_{t_0}^{t_1} \nu \|\nabla u(s)\|^2 ds = 0.$$

Opening the first integral by the fundamental theorem of calculus, we get (6.1). $\square$

Finally, we are ready for the second key result of this paper.

THEOREM 6.6 (Ladyzhenskaya: global strong solutions in 2D [8]). *Let $u_0 \in V(\mathbb{T}^2)$. Then, u_0 gives rise to a strong global-in-time solution of the Navier–Stokes equations.*

Proof. We argue by an a priori estimate on the Galerkin approximations followed by continuation, so that no strong solution is assumed to exist in advance.

Every step below is carried out for the Galerkin approximation u_n , which lives in the finite-dimensional space $P_n H$ where all the pairings are classical and $Au_n \in P_n H \subset \mathbb{L}^2$ automatically. The resulting bound is uniform in n , so it passes to the limit constructed in Theorem 5.3 by the same weak lower semicontinuity used there, and Lemma 6.3 then applies to the limit. To keep the notation light we write u for u_n throughout the computation and restore the subscript at the end.

Taking the inner product of the Galerkin equation with Au :

$$\langle \partial_t u, Au \rangle + \nu \langle Au, Au \rangle + \langle (u \cdot \nabla)u, Au \rangle = 0$$

Since $\langle \partial_t u, Au \rangle = \langle \partial_t u, -\Delta u \rangle = \langle \nabla \partial_t u, \nabla u \rangle$ on the torus, we can transform the first term:

$$\langle \partial_t u, Au \rangle = \frac{1}{2} \frac{d}{dt} \|\nabla u\|^2$$

Which leads to the following expression:

$$(6.2) \quad \frac{1}{2} \frac{d}{dt} \|\nabla u\|^2 + \nu \|Au\|^2 = -\langle (u \cdot \nabla)u, Au \rangle$$

We bound the term on the right-hand side with Cauchy–Schwarz inequality:

$$|\langle (u \cdot \nabla)u, Au \rangle| \leq \|(u \cdot \nabla)u\|_{\mathbb{L}^2} \|Au\|_{\mathbb{L}^2}$$

Using Hölder inequality with exponents $(4, 4)$ since $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$:

$$\|(u \cdot \nabla)u\|_{\mathbb{L}^2} \leq \|u\|_{\mathbb{L}^4} \|\nabla u\|_{\mathbb{L}^4}$$

Now we use Theorem 2.18 to control the $\mathbb{L}^4$ -normed terms:

$$\begin{aligned} \|u\|_{\mathbb{L}^4} &\leq C_1 \|u\|_{\mathbb{L}^2}^{1/2} \|\nabla u\|_{\mathbb{L}^2}^{1/2} \\ \|\nabla u\|_{\mathbb{L}^4} &\leq C_2 \|\nabla u\|_{\mathbb{L}^2}^{1/2} \|\Delta u\|_{\mathbb{L}^2}^{1/2} \end{aligned}$$

Returning the result back into the original expression:

$$\langle (u \cdot \nabla)u, Au \rangle \leq C \|u\|_{\mathbb{L}^2}^{1/2} \|\nabla u\|_{\mathbb{L}^2}^{1/2} \|\nabla u\|_{\mathbb{L}^2}^{1/2} \|\Delta u\|_{\mathbb{L}^2}^{1/2} \|Au\|_{\mathbb{L}^2}$$

Considering that $\|\Delta u\|_{\mathbb{L}^2} = \|Au\|_{\mathbb{L}^2}$, several terms collapse:

$$\langle (u \cdot \nabla)u, Au \rangle \leq C \|u\|_{\mathbb{L}^2}^{1/2} \|\nabla u\|_{\mathbb{L}^2} \|Au\|_{\mathbb{L}^2}^{3/2}$$

Returning to (6.2):

$$\frac{1}{2} \frac{d}{dt} \|\nabla u\|^2 + \nu \|Au\|^2 \leq C \|u\|_{\mathbb{L}^2}^{1/2} \|\nabla u\|_{\mathbb{L}^2} \|Au\|_{\mathbb{L}^2}^{3/2}$$

We use Young's inequality to carefully handle $\|Au\|_{\mathbb{L}^2}^{3/2}$. Let $a = \|u\|_{\mathbb{L}^2}^{1/2} \|\nabla u\|_{\mathbb{L}^2}$ and $b = \|Au\|_{\mathbb{L}^2}^{3/2}$. We know that

$$ab \leq \frac{\epsilon^{1-p}}{p} a^p + \frac{\epsilon}{q} b^q$$

with $\frac{1}{p} + \frac{1}{q} = 1$ and $p, q > 1$. Apply this to the whole right-hand side Cab , taking $p = 4$, $q = 4/3$ and $\epsilon = \frac{2\nu}{3C}$, so that the b -term becomes $C \cdot \frac{3}{4} \cdot \frac{2\nu}{3C} \|Au\|^2 = \frac{\nu}{2} \|Au\|^2$:

$$C\|u\|_{\mathbb{L}^2}^{1/2} \|\nabla u\|_{\mathbb{L}^2} \|Au\|_{\mathbb{L}^2}^{3/2} \leq \frac{C_3}{\nu^3} \|u\|_{\mathbb{L}^2}^2 \|\nabla u\|_{\mathbb{L}^2}^4 + \frac{\nu}{2} \|Au\|_{\mathbb{L}^2}^2$$

where $C_3 = \frac{1}{4} \left(\frac{3C}{2\nu}\right)^3 \nu^3 = \frac{27C^3}{32}$ absorbs the numerical factors. Pasting to (6.2):

$$\frac{1}{2} \frac{d}{dt} \|\nabla u\|^2 + \nu \|Au\|^2 \leq \frac{C_3}{\nu^3} \|u\|_{\mathbb{L}^2}^2 \|\nabla u\|_{\mathbb{L}^2}^4 + \frac{\nu}{2} \|Au\|_{\mathbb{L}^2}^2$$

Subtracting $\frac{\nu}{2} \|Au\|^2$ from both sides absorbs that term into the left:

$$\frac{1}{2} \frac{d}{dt} \|\nabla u\|^2 + \frac{\nu}{2} \|Au\|^2 \leq \frac{C_3}{\nu^3} \|u\|_{\mathbb{L}^2}^2 \|\nabla u\|_{\mathbb{L}^2}^4$$

The quantity $\frac{\nu}{2} \|Au\|^2$ is non-negative, so discarding it only weakens the inequality and yields a differential inequality:

$$\frac{d}{dt} \|\nabla u\|^2 \leq \frac{2C_3}{\nu^3} \|u\|_{\mathbb{L}^2}^2 \|\nabla u\|_{\mathbb{L}^2}^4$$

Transforming the expression:

$$(6.3) \quad \frac{d}{dt} \|\nabla u\|^2 \leq \left(\frac{2C_3}{\nu^3} \|u\|_{\mathbb{L}^2}^2 \|\nabla u\|_{\mathbb{L}^2}^2 \right) \|\nabla u\|^2$$

Let $g(t) = \frac{2C_3}{\nu^3} \|u\|_{\mathbb{L}^2}^2 \|\nabla u\|_{\mathbb{L}^2}^2$. Since we are working with $u = u_n$, Lemma 4.4 bounds $\|u_n(t)\|_{\mathbb{L}^2}^2$ by $\|u_0\|_{\mathbb{L}^2}^2$ uniformly in t and n , and Lemma 4.5 bounds $\int_0^T \|\nabla u_n\|_{\mathbb{L}^2}^2 dt$ by $\frac{1}{2\nu} \|u_0\|_{\mathbb{L}^2}^2$. So g is a uniformly bounded factor times a factor that is integrable in time — that is, $\|\nabla u_n\|^2$ lies in $L^1(0, T)$, being the square of an $L^2(0, T)$ quantity — whence $g \in L^1([0, T])$ with a bound independent of n . Let $\phi(t) = \|\nabla u_n(t)\|^2$, which is C^1 in t because u_n solves a finite-dimensional ODE with smooth coefficients.

Since we operate on a periodic torus $\mathbb{T}^2$, the Poincaré inequality makes $\|\cdot\|_{\mathbb{H}^1}$ and $\|\nabla \cdot\|_{\mathbb{L}^2}$ equivalent norms on V , so this is the same as absolute continuity of $t \mapsto \|u_n(t)\|_{\mathbb{H}^1}^2$. With both hypotheses of Lemma 2.3 satisfied, we can invoke it to get:

$$(6.4) \quad \|\nabla u_n(t)\|^2 \leq \|\nabla P_n u_0\|^2 \exp \left(\int_0^t g(s) ds \right) \leq \|\nabla u_0\|^2 \exp \left(\int_0^T g(s) ds \right)$$

for all $t \in [0, T]$, using $\|\nabla P_n u_0\| \leq \|\nabla u_0\|$ from Lemma 4.1, which is finite precisely because $u_0 \in V$. The right-hand side is a constant $M(T, \nu, \|u_0\|_V)$ independent of n .

Feeding this back into the inequality before we discarded the dissipation term and integrating on $[0, T]$ also gives $\nu \int_0^T \|Au_n\|^2 dt \leq M'$, again uniformly in n . Hence (u_n) is bounded in $L^\infty(0, T; V) \cap L^2(0, T; D(A))$, and by weak lower semicontinuity the limit u of Theorem 5.3 inherits both bounds:

$$u \in L^\infty(0, T; \mathbb{H}^1) \cap L^2(0, T; \mathbb{H}^2),$$

where $\|Au\|_{\mathbb{L}^2}$ controls $\|u\|_{\mathbb{H}^2}$ by Theorem 3.3. This is exactly Definition 6.1, so u is a strong solution on $[0, T]$. Since M is finite for every $T > 0$ and the estimate never degenerates, no finite time is distinguished, and the solution is global in time. $\square$

We close the two-dimensional theory with its last main result, which is where the restriction to two dimensions does the decisive work.

THEOREM 6.7 (Uniqueness of 2D Leray–Hopf weak solutions). *Let $u_0 \in H(\mathbb{T}^2)$. Then the Leray–Hopf weak solution of the Navier–Stokes equations on $\mathbb{T}^2$ with initial datum u_0 is unique. In particular strong solutions, being Leray–Hopf weak solutions, are unique.*

Proof. Let u and v be two Leray–Hopf weak solutions starting from the exact same initial condition $u(0) = v(0) = u_0$. We denote their difference as $w = u - v$; since they start from the same initial condition, $w(0) = 0$.

Both lie in $L^\infty(0, T; H) \cap L^2(0, T; V)$ by Definition 3.7, so w does too, and the bound on ∂_t established for the approximations transfers to each of them, giving $\partial_t w \in L^2(0, T; V')$. Theorem 2.17 therefore applies to w with the Gelfand triple $V \subset H \subset V'$: $w \in C([0, T]; H)$, the map $t \mapsto \|w(t)\|^2$ is absolutely continuous, and $\frac{d}{dt}\|w\|^2 = 2\langle \partial_t w, w \rangle_{V', V}$. This is what licenses the computation below in the weak setting, where $\partial_t w$ need not be an $\mathbb{L}^2$ function. Note that only $\nabla u \in L^2(0, T; \mathbb{L}^2)$ will be used, which every Leray–Hopf weak solution supplies; the additional regularity of a strong solution is not needed.

We take the difference of the two formulations for u and v to yield the formulation for w :

$$\partial_t(u - v) + \nu A(u - v) + (u \cdot \nabla)u - (v \cdot \nabla)v = 0$$

Similarly to (5.5):

$$(u \cdot \nabla)u - (v \cdot \nabla)v = ((u - v) \cdot \nabla)u + (u \cdot \nabla)(u - v)$$

Which yields:

$$\partial_t w + \nu A w + (w \cdot \nabla)u + (u \cdot \nabla)w = 0$$

Taking the $\mathbb{L}^2$ inner product with w :

$$\frac{1}{2} \frac{d}{dt} \|w\|^2 + \nu \|\nabla w\|^2 + \langle (w \cdot \nabla)u, w \rangle + \langle (u \cdot \nabla)w, w \rangle = 0$$

By Lemma 3.6, $\langle (u \cdot \nabla)w, w \rangle = 0$. That results in:

$$\frac{1}{2} \frac{d}{dt} \|w\|^2 + \nu \|\nabla w\|^2 = -\langle (w \cdot \nabla)u, w \rangle$$

We bound the right side by Hölder’s inequality with exponents $(4, 2, 4)$:

$$\langle (w \cdot \nabla)u, w \rangle \leq |\langle (w \cdot \nabla)u, w \rangle| \leq \|w\|_{\mathbb{L}^4} \|\nabla u\|_{\mathbb{L}^2} \|w\|_{\mathbb{L}^4} = \|w\|_{\mathbb{L}^4}^2 \|\nabla u\|_{\mathbb{L}^2}$$

Applying Theorem 2.18:

$$\|w\|_{\mathbb{L}^4}^2 \|\nabla u\|_{\mathbb{L}^2} \leq C \|w\|_{\mathbb{L}^2} \|\nabla w\|_{\mathbb{L}^2} \|\nabla u\|_{\mathbb{L}^2}$$

Using Peter–Paul inequality with $a = \|w\|_{\mathbb{L}^2} \|\nabla u\|_{\mathbb{L}^2}$, $b = \|\nabla w\|_{\mathbb{L}^2}$, $\epsilon = \frac{\nu}{2C}$:

$$C \|w\|_{\mathbb{L}^2} \|\nabla w\|_{\mathbb{L}^2} \|\nabla u\|_{\mathbb{L}^2} \leq \frac{C}{\nu} \|w\|_{\mathbb{L}^2}^2 \|\nabla u\|_{\mathbb{L}^2}^2 + \frac{\nu}{2} \|\nabla w\|_{\mathbb{L}^2}^2$$

Term $\frac{\nu}{2}\|\nabla w\|_{\mathbb{L}^2}^2$ is absorbed, leaving:

$$\frac{1}{2}\frac{d}{dt}\|w\|^2 + \frac{\nu}{2}\|\nabla w\|^2 \leq \frac{C}{\nu}\|w\|_{\mathbb{L}^2}^2\|\nabla u\|_{\mathbb{L}^2}^2$$

We adjust the expression to yield a differential inequality:

$$\frac{1}{2}\frac{d}{dt}\|w\|^2 + \frac{\nu}{2}\|\nabla w\|^2 \leq \left(\frac{C}{\nu}\|\nabla u\|_{\mathbb{L}^2}^2\right)\|w\|_{\mathbb{L}^2}^2$$

The term $\frac{\nu}{2}\|\nabla w\|^2$ is non-negative, so discarding it only weakens the inequality; multiplying by 2 we get:

$$\frac{d}{dt}\|w\|^2 \leq \left(\frac{2C}{\nu}\|\nabla u\|_{\mathbb{L}^2}^2\right)\|w\|^2$$

Let $g(t) = \frac{2C}{\nu}\|\nabla u(t)\|_{\mathbb{L}^2}^2$. This lies in $L^1([0, T])$ because $u \in L^2(0, T; V)$ by Definition 3.7(i) — the square of an L^2 -in-time quantity is L^1 in time. Absolute continuity of $t \mapsto \|w(t)\|^2$ was established at the start of the proof.

Thus, we use Lemma 2.3 to yield:

$$\|w(t)\|^2 \leq \|w(0)\|^2 \exp\left(\int_0^t g(s)ds\right)$$

But since $w(0) = 0$, the right-hand side is 0, and $\|w(t)\|^2 \geq 0$; hence $w(t) = 0$ for every $t \in [0, T]$. As $T > 0$ was arbitrary, $u = v$ for all time. $\square$

7. Existence of strong solutions in 3D. In this section we work on the periodic torus $\mathbb{T}^3$ instead of $\mathbb{T}^2$. The Galerkin and compactness strategy of section 3 through section 5 also yields global Leray–Hopf weak solutions in 3D, but not verbatim. The Galerkin construction and the energy estimate do not depend on the dimension; the bound on the time derivative that feeds Theorem 5.2, however, was obtained from two-dimensional interpolation. On $\mathbb{T}^3$ the corresponding inequality, Theorem 7.1 below, gives $\|u_n\|_{\mathbb{L}^4}^2 \leq C\|u_n\|_{\mathbb{L}^2}^{1/2}\|\nabla u_n\|_{\mathbb{L}^2}^{3/2}$, and the same computation then bounds $\partial_t u_n$ only in $L^{4/3}(0, T; V'_3)$, with V'_3 as in Definition 7.2, rather than in L^2 . This still suffices, because Theorem 5.2 requires only an exponent $p > 1$. What does not carry over is the argument of section 6 that upgrades weak solutions to strong ones, and the purpose of this section is to locate precisely where that argument fails.

We begin with a restated Ladyzhenskaya inequality for $\mathbb{T}^3$.

THEOREM 7.1 (Ladyzhenskaya, 3D case [8, 11]). *On the periodic torus $\mathbb{T}^3$, for $u \in \dot{\mathbb{H}}^1(\mathbb{T}^3)$, which provides the zero-mean condition, we have the following:*

$$\|u\|_{\mathbb{L}^4} \leq C\|u\|_{\mathbb{L}^2}^{1/4}\|\nabla u\|_{\mathbb{L}^2}^{3/4}$$

We state this without proof; it is the case $n = 3$ of the Gagliardo–Nirenberg interpolation inequality, and the argument runs parallel to Theorem 2.18 with the slice bound applied in three variables rather than two [11]. The exponents are what matter here: the weight has shifted from $\frac{1}{2}, \frac{1}{2}$ to $\frac{1}{4}, \frac{3}{4}$, putting more of the burden on the gradient. As we shall see, the Ladyzhenskaya inequality in 3D no longer allows for the estimates to be closed the same way as they were closed in the 2-dimensional case.

In addition to that, we will take the curl of the Navier–Stokes equations to examine the vorticity propagation.

For the 2D case:

$$\begin{aligned}\nabla \times (\partial_t u + (u \cdot \nabla)u) &= \nabla \times \left(-\nabla p + \nu \Delta u \right) \\ \partial_t \omega + (u \cdot \nabla)\omega &= 0 + \nu \Delta \omega \\ \partial_t \omega + (u \cdot \nabla)\omega &= \nu \Delta \omega\end{aligned}$$

In two dimensions ω is a scalar and the stretching term $(\omega \cdot \nabla)u$ is absent, so vorticity is merely transported and diffused; it cannot be amplified by the velocity gradient. However, in 3 dimensions, the curl of the Navier–Stokes equations takes another shape.

In 3 dimensions, we re-open the expression except for the vorticity term the same way:

$$\begin{aligned}\nabla \times (\partial_t u + (u \cdot \nabla)u) &= \nabla \times \left(-\nabla p + \nu \Delta u \right) \\ \partial_t \omega + \nabla \times ((u \cdot \nabla)u) &= \nu \Delta \omega\end{aligned}$$

We know that by Lamb vector decomposition:

$$(u \cdot \nabla)u = \nabla \left(\frac{1}{2} |u|^2 \right) - u \times \omega$$

Returning the decomposition:

$$\begin{aligned}\partial_t \omega + \nabla \times \left(\nabla \left(\frac{1}{2} |u|^2 \right) - u \times \omega \right) &= \nu \Delta \omega \\ \partial_t \omega - \nabla \times (u \times \omega) &= \nu \Delta \omega\end{aligned}$$

Opening the cross product by the corresponding vector calculus identity:

$$\nabla \times (u \times \omega) = u(\nabla \cdot \omega) - \omega(\nabla \cdot u) + (\omega \cdot \nabla)u - (u \cdot \nabla)\omega$$

Returning it:

$$\partial_t \omega - \left(u(\nabla \cdot \omega) - \omega(\nabla \cdot u) + (\omega \cdot \nabla)u - (u \cdot \nabla)\omega \right) = \nu \Delta \omega$$

$\nabla \cdot u = 0$ by incompressibility; additionally, $\nabla \cdot \omega = \nabla \cdot (\nabla \times u) = 0$. Thus, we are only left with:

$$\begin{aligned}\partial_t \omega - \left((\omega \cdot \nabla)u - (u \cdot \nabla)\omega \right) &= \nu \Delta \omega \\ \partial_t \omega - (\omega \cdot \nabla)u + (u \cdot \nabla)\omega &= \nu \Delta \omega \\ \partial_t \omega + (u \cdot \nabla)\omega &= \nu \Delta \omega + (\omega \cdot \nabla)u\end{aligned}$$

The term $(\omega \cdot \nabla)u$, absent in the 2D vorticity equation but present in 3D vorticity equation, represents the vortex stretching happening in 3 dimensions, with velocity gradient ∇u pulling at ω and allowing the vortex to twist and stretch.

Now, suppose the enstrophy estimate will close the same way regardless of the changed vorticity equation. To proceed with this, we first define a functional space V_3 analogous to Definition 2.15 for $\mathbb{T}^3$.

DEFINITION 7.2. *We define the space of time-independent test functions*

$$V_3 = \{u \in \dot{\mathbb{H}}^1(\mathbb{T}^3) : k \cdot \hat{u}_k = 0 \ \forall k\}$$

with norm given as $\|\nabla u\|_{\mathbb{L}^2}$. We additionally define V'_3 as:

$$V'_3 = \{f : k \cdot \hat{f}_k = 0, \sum_{k \neq 0} |k|^{-2} |\hat{f}_k|^2 < \infty\}.$$

with a norm given as:

$$\|f\|_{V'_3} = \sup_{\|x\|_{V_3} \leq 1} |\langle f, x \rangle|$$

Here $H_3 = H(\mathbb{T}^3)$ denotes the divergence-free, mean-zero subspace of $\dot{\mathbb{L}}^2(\mathbb{T}^3)$, defined exactly as in Definition 2.10 with $\dot{\mathbb{Z}}^3$ in place of $\dot{\mathbb{Z}}^2$. By the Riesz representation theorem we identify H_3 with its dual H'_3 , so that

$$\langle f, v \rangle_{H'_3, H_3} = \langle f, v \rangle_{\mathbb{L}^2}.$$

$V_3 \subset H_3$ densely and continuously, hence $H_3 \cong H'_3 \subset V'_3$, and the $V'_3 - V_3$ pairing extends the $\mathbb{L}^2$ inner product. We do not identify V_3 with V'_3 .

Now, we are ready to formalize our claim. We state it as a hypothesis rather than a theorem, because what follows is an attempt that does not succeed; the argument below is presented in order to exhibit the exact point of failure, and it is deliberately not closed by a proof.

Hypothesis 7.3. Similarly to Theorem 6.6, every $u_0 \in V_3(\mathbb{T}^3)$ gives rise to a global-in-time strong solution.

Attempted proof. Let us assume the hypothesis is true. Then, we should be able to control the expression

$$(7.1) \quad \partial_t u + \nu Au + (u \cdot \nabla)u = 0$$

in the Lebesgue space $\mathbb{L}^2$, since Definition 6.1 requires $u \in L^\infty(0, T; \mathbb{H}^1) \cap L^2(0, T; \mathbb{H}^2)$.

We assume that the local strong solution u on $[0, T]$ exists and $u \in L^\infty(0, T; \mathbb{H}^1) \cap L^2(0, T; \mathbb{H}^2)$ and all terms present in (7.1) belong to $L^2(0, T; \mathbb{L}^2)$, analogous to the proof of Theorem 6.6 (we will return to examine this assumption later). Taking the inner product with $Au \in \mathbb{L}^2$, we get:

$$\langle \partial_t u, Au \rangle + \nu \langle Au, Au \rangle + \langle (u \cdot \nabla)u, Au \rangle = 0$$

Under the assumptions above, this yields:

$$\frac{1}{2} \frac{d}{dt} \|\nabla u\|^2 + \nu \|Au\|^2 = -\langle (u \cdot \nabla)u, Au \rangle$$

Using Theorem 7.1 and Hölder inequality with exponents $(4, 4, 2)$, we attempt to

bound the pairing of the nonlinear term with Au :

$$\begin{aligned}\langle (u \cdot \nabla)u, Au \rangle &\leq \|u\|_{\mathbb{L}^4} \|\nabla u\|_{\mathbb{L}^4} \|Au\|_{\mathbb{L}^2} \\ \|u\|_{\mathbb{L}^4} &\leq C_1 \|u\|_{\mathbb{L}^2}^{1/4} \|\nabla u\|_{\mathbb{L}^2}^{3/4} \\ \|\nabla u\|_{\mathbb{L}^4} &\leq C_2 \|\nabla u\|_{\mathbb{L}^2}^{1/4} \|Au\|_{\mathbb{L}^2}^{3/4} \\ \langle (u \cdot \nabla)u, Au \rangle &\leq C \|u\|_{\mathbb{L}^2}^{1/4} \|\nabla u\|_{\mathbb{L}^2} \|Au\|_{\mathbb{L}^2}^{7/4}\end{aligned}$$

We invoke Young's inequality on the whole right-hand side Cab and set $a = \|u\|_{\mathbb{L}^2}^{1/4} \|\nabla u\|_{\mathbb{L}^2}$, $b = \|Au\|_{\mathbb{L}^2}^{7/4}$, $p = 8$, $q = \frac{8}{7}$, $\epsilon = \frac{4\nu}{7C}$, so that the b -term is again $\frac{\nu}{2} \|Au\|^2$:

$$C \|u\|_{\mathbb{L}^2}^{1/4} \|\nabla u\|_{\mathbb{L}^2} \|Au\|_{\mathbb{L}^2}^{7/4} \leq \frac{C'}{\nu^7} \|u\|_{\mathbb{L}^2}^2 \|\nabla u\|_{\mathbb{L}^2}^8 + \frac{\nu}{2} \|Au\|_{\mathbb{L}^2}^2$$

Absorbing the term with the Stokes operator on the right-hand side, we get:

$$\frac{1}{2} \frac{d}{dt} \|\nabla u\|^2 + \frac{\nu}{2} \|Au\|^2 \leq \frac{C'}{\nu^7} \|u\|_{\mathbb{L}^2}^2 \|\nabla u\|_{\mathbb{L}^2}^8$$

To set up the differential inequality, we discard the non-negative term $\frac{\nu}{2} \|Au\|^2$ and multiply by 2, getting:

$$\frac{d}{dt} \|\nabla u\|^2 \leq \left(\frac{2C'}{\nu^7} \|u\|_{\mathbb{L}^2}^2 \|\nabla u\|_{\mathbb{L}^2}^6 \right) \|\nabla u\|^2$$

Compare this with (6.3). There the bracket was $\|u\|^2 \|\nabla u\|^2$, and the energy inequality gave exactly what was needed: $\|u\|^2$ bounded uniformly and $\|\nabla u\|^2$ integrable in time, so the bracket was in $L^1(0, T)$ and Grönwall closed the estimate. Here the bracket is $\|u\|^2 \|\nabla u\|^6$. The energy inequality still gives $\|\nabla u\|^2 \in L^1(0, T)$, but that says nothing about $\|\nabla u\|^6$; membership of $\|\nabla u\|^6$ in $L^1(0, T)$ would require $\nabla u \in L^6(0, T; \mathbb{L}^2)$, which is strictly more than the energy estimate provides. Thus, we cannot claim that

$$g(t) = \frac{2C'}{\nu^7} \|u\|_{\mathbb{L}^2}^2 \|\nabla u\|_{\mathbb{L}^2}^6$$

is $L^1([0, T])$, and Lemma 2.3 does not apply. The argument stops here; it is not completed.

This breakdown is not a sleight of hand, but it is a limitation of this particular estimate rather than a statement about the equations. By the Fujita–Kato theorem [6], a local-in-time strong solution exists on a positive interval, so the assumption made at the start of the attempted proof is legitimate there. The estimate above does not provide an a priori bound sufficient to continue that solution globally. It therefore does not establish global-in-time strong existence; it also does not establish finite-time breakdown.

8. Conclusion. The arguments given here for the 2-dimensional Navier–Stokes problem are stated on the periodic torus $\mathbb{T}^2$, and two of the steps genuinely use that choice. The Aubin–Lions argument of section 5 relies on the compactness of the embedding $\mathbb{H}^1 \hookrightarrow \mathbb{L}^2$, which holds on the torus but fails on $\mathbb{R}^2$, where mass can escape to infinity; and the Galerkin basis of section 4 is available in closed form only because

the Stokes eigenfunctions are the divergence-free Fourier modes. The corresponding results on $\mathbb{R}^2$ and on bounded domains are true, but they are not corollaries of what is written above: recovering them requires local compactness arguments in place of Theorem 5.2, and on a domain with boundary, the trace theory that the torus was chosen to avoid [12, 11].

In three dimensions the energy inequality yields $u \in L^\infty(0, T; H_3) \cap L^2(0, T; V_3)$. It controls the $\mathbb{L}^2$ norm uniformly in time and the time integral of $\|\nabla u\|_{\mathbb{L}^2}^2$, but it does not control the higher powers of $\|\nabla u\|_{\mathbb{L}^2}$ that enter the Grönwall coefficient of section 7; consequently, the argument given here does not establish global strong regularity. A familiar heuristic places this in context. On $\mathbb{R}^3$ the Navier–Stokes equations are invariant under the rescaling $u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t)$, which leaves the homogeneous norm of $\dot{H}^{1/2}(\mathbb{R}^3)$ unchanged; both quantities controlled by the energy inequality sit half a derivative below that scaling-critical level, which is the sense of the phrase “half-derivative gap”. This comparison is useful context for why the three-dimensional estimate does not close, but it is a statement about $\mathbb{R}^3$ and is not itself a proof of an obstruction on the periodic torus. We make no claims about the Navier–Stokes existence and smoothness problem beyond what is proved above.

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